

Class 05

경험_부작용(멀미)

Hyungseok Kim
Konkuk University





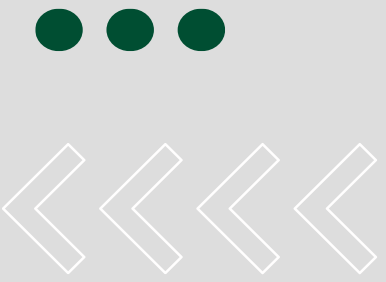
VR Health Effects

Measuring VR Sickness and Presence

HyungSeok Kim
Konkuk University



VR 멀미 개요 멀미부터 인체 공학까지



01.건강 영향

신체가 가상 환경에
반응하는 방식

Health Effects

02.멀미

이론적 배경, 유형 구분
및 원인

Motion Sickness

03.기타건강이슈

이론적 배경, 유형 구분
및 원인

Other Health Issues

VR 멀미의 유형 구분 및 정의



Motion Sickness

Illness arising from physical motion environments (e.g., cars, boats).



Visually Induced Motion Sickness (VIMS)

Sickness from visual motion contexts alone, without physical movement.



Simulator Sickness

Sickness arising specifically from simulator contexts.



Cybersickness

Sickness specifically occurring after exposure to Virtual Reality (VR).

What is VR Sickness?



Virtual Reality sickness occurs when exposure to a virtual environment causes symptoms that are similar to **motion sickness symptoms**.

Motion sickness refers to **illness arising** from a person being within a motion environment, such as a moving vehicle.

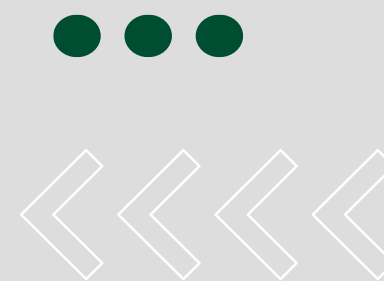
신체 감각의 충돌과 생존 본능



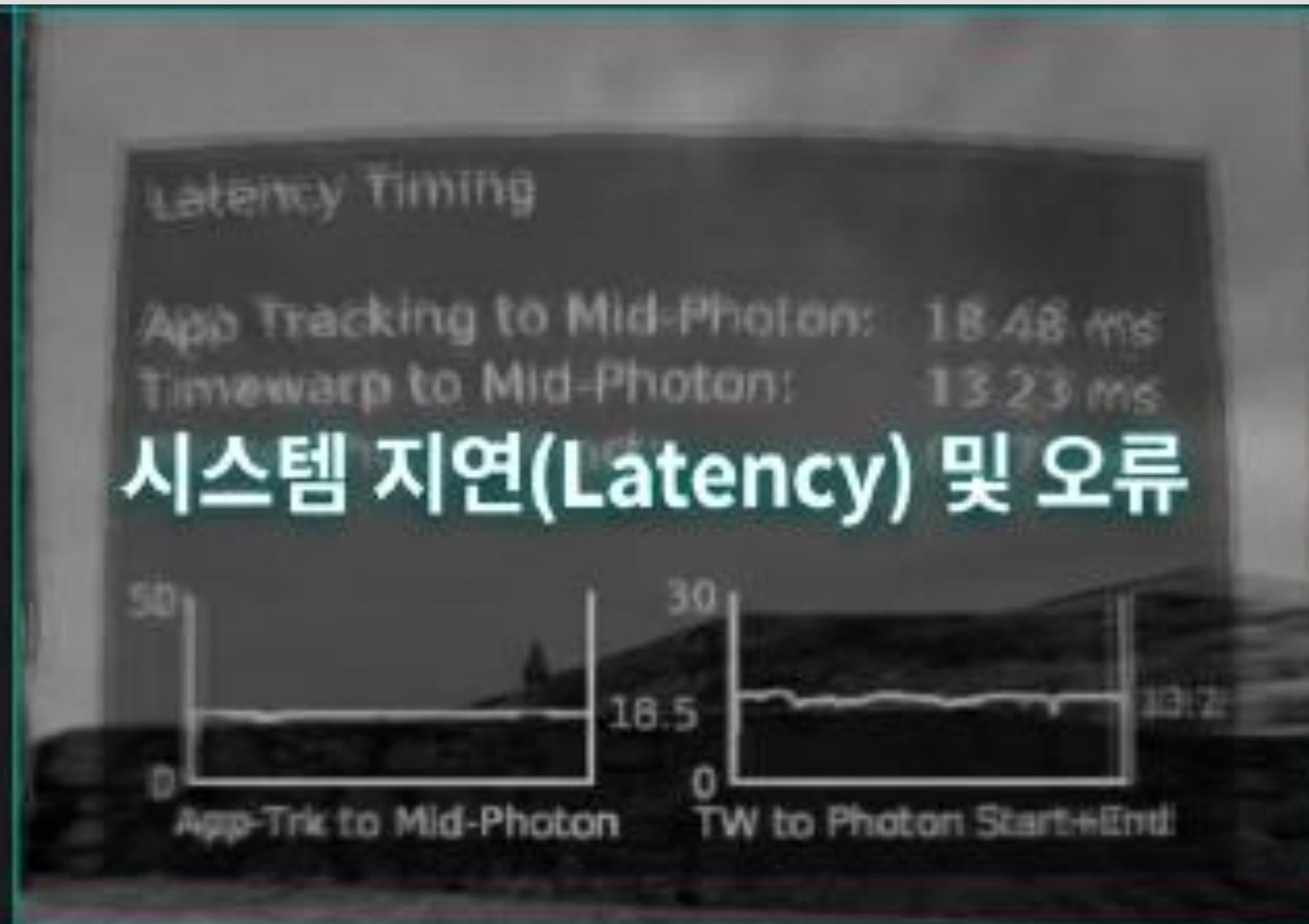
Evolutionary Theory: The body ejects perceived poisons.

시각적 움직임과 지연 시간의 영향

Scene Motion : 현실과 다른 가상 환경 전체의 시각적 움직임



현실 세계의 물리적 움직임



의도적인 장면 움직임 (Intentional)

콘텐츠 연출을 위한 카메라 이동.

비의도적인 장면 움직임 (Unintentional)

트래킹 오류로 인한 화면 밀림. 심각한 감각 갈등 유발.

멀미 증상의 정량적 측정 요소



강도 (Intensity)

증상이 얼마나 심각한가?



발현 속도 (Rate of Onset)

자극 제시 후 증상이 얼마나 빠르게 나타나는가?



회복 속도 (Rate of Decay)

자극 제거 후 증상이 얼마나 빠르게 감소하는가?



사용자 비율 (Percentage of Users)

특정 수준 이상의 증상을 겪는 사용자 비율.

Questionnaires

Simulator sickness questionnaires (SSQ)

- The most common questionnaire to evaluate the simulator sickness.
- Measuring motion sickness when using a simulator for virtual environments.
- It may impact the effects on Head-mounted displays.

SIMULATOR SICKNESS QUESTIONNAIRE

Kennedy, Lane, Berbaum, & Lilenthal (1993)***

Instructions : Circle how much each symptom below is affecting you right now.

1. General discomfort	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
2. Fatigue	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
3. Headache	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
4. Eye strain	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
5. Difficulty focusing	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
6. Salivation increasing	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
7. Sweating	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
8. Nausea	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
9. Difficulty concentrating	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
10. Fullness of the head	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
11. Blurred vision	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
12. Dizziness with eyes open	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
13. Dizziness with eyes closed	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
14. *Vertigo	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
15. **Stomach awareness	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>
16. Burping	<u>None</u>	<u>Slight</u>	<u>Moderate</u>	<u>Severe</u>

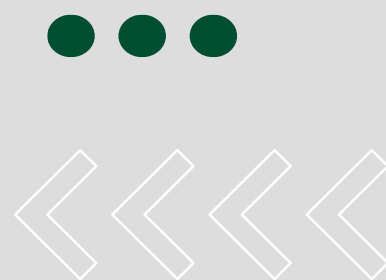
* Vertigo is experienced as loss of orientation with respect to vertical upright.

** Stomach awareness is usually used to indicate a feeling of discomfort which is just short of nausea.

Last version : March 2013

***Original version : Kennedy, R.S., Lane, N.E., Berbaum, K.S., & Lilenthal, M.G. (1993). Simulator Sickness Questionnaire: An enhanced method for quantifying simulator sickness. *International Journal of Aviation Psychology*, 3(3), 203-220.

VR motion sickness questionnaires



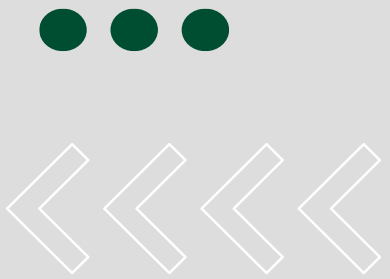
- A questionnaire about motion sickness in virtual reality environment was developed.
- The existing simulator sickness questionnaire was modified to develop the virtual reality sickness questionnaire (VRSQ).
- The newly derived VRSQ consisted of 9 items, which are included in two components, namely oculomotor and disorientation.

Table 7. Virtual reality sickness questionnaire (VRSQ).

VRSQ symptom	Oculomotor	Disorientation
1. General discomfort	0	
2. Fatigue	0	
3. Eyestrain	0	
4. Difficulty focusing	0	
5. Headache		0
6. Fullness of head		0
7. Blurred vision		0
8. Dizzy (eyes closed)		0
9. Vertigo		0
Total	[1]	[2]

주관적 평가 도구 : VRSQ

VRSQ : VR Sickness Questionnaire



Modified from SSQ for VR environments

9 items from original SSQ (16 items)

Two components: Oculomotor and Disorientation

More focused on VR-specific symptoms

Simpler administration and scoring

Virtual Reality Sickness Questionnaire(Kim et al.,2018)

가상현실 환경 전용으로 개발된 표준화 설문지

기존 SSQ의 한계를 극복.

HMD하드웨어 및 콘텐츠 사용성 평가의 핵심 지표

VR의 특수성을 반영하여 정확한 멀미 정도를 산출한다.

주관적 평가 도구 : VRSQ

VRSQ : VR Sickness Questionnaire



Legacy SSQ

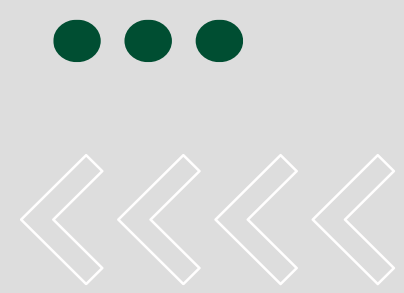


Legacy SSQ
(16 Items)



VRSQ (9 Items)	
 Oculomotor (안구 운동)	 Disorientation (방향 감각 상실)
• General discomfort (일반적인 불편함)	• Headache (두통)
• Fatigue (피로)	• Fullness of head (머리가 꽉 찬 느낌)
• Eyestrain (눈의 피로)	• Blurred vision (시야 흐림)
• Difficulty focusing (초점 맞추기 어려움)	• Dizziness (eyes closed) (어지러움, 눈 감았을 때)
	• Vertigo (현기증)
Total (합계) [1]	[2]

Motion Sickness Assessment Questionnaire (MSAQ)



APPENDIX A
MOTION SICKNESS ASSESSMENT QUESTIONNAIRE (MSAQ).

Instructions. Using the scale below, please rate how accurately the following statements describe your experience

Not at all

Severely

1

2

3

4

5

6

7

8

9

1. I felt sick to my stomach (G)

2. I felt faint-like (C)

3. I felt annoyed/irritated (S)

4. I felt sweaty (P)

5. I felt queasy (G)

6. I felt lightheaded (C)

7. I felt drowsy (S)

8. I felt clammy/cold sweat (P)

9. I felt disoriented (Q)

10. I felt tired/fatigued (S)

11. I felt nauseated (G)

12. I felt hot/warm (P)

13. I felt dizzy (C)

14. I felt like I was spinning (C)

15. I felt as if I may vomit (G)

16. I felt uneasy (S)

Note. G; Gastrointestinal; C: Central; P: Peripheral; SR; Somite-related.

The overall motion sickness score is obtained by calculating the percentage of total points scored: (sum of points from all items/144) × 100. Subscale scores are obtained by calculating the percent of points scored within each factor: (sum of gastrointestinal items/36) × 100; (sum of central items/45) × 100; (sum of peripheral items/27) × 100; (sum of somite-related items/36) × 100.

Physiological measurements : ECG & EGG



ECG (심전도)

심박수(Heart Rate) 증가 = 멀미 발생 징후.



심박수(Heart Rate) 증가 = 멀미 발생 징후.

EGG (위전도)



위장 불편감(Gastric Dysrhythmia) 측정.



객관적 데이터 (Objective Data)

Physiological measurements



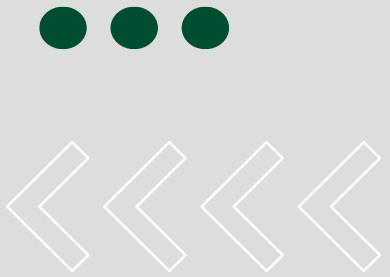
- Electrocardiogram (ECG) records the electrical activity of the heart by placing electrodes on the skin. Heart rate typically increases during a VR session.
- Electrogastrogram (EGG) is similar to the ECG, but the electrodes are placed near the stomach so that gastrointestinal discomfort can be estimated.
- Electrooculogram (EOG) are placed around the eyes so that eye movement can be estimated. Alternatively, a camera-based eye tracking system may be used.
- Photoplethysmogram (PPG) provides additional data on heart movement and is obtained by using a pulse oximeter.
- Galvanic skin response (GSR) measures electrical resistance across the surface of the skin.

Physiological measurements



- Respiratory effort: The breathing rate and amplitude are measured from a patch on the chest that responds to differential pressure or expansion. The rate of breathing may increase during the VR session.
- Skin pallor: This can be measured using a camera and image processing. In the simplest case, an IR LED and photodiode serves as an emitter-detector pair that measures skin reflectance.
- Head motion: A head tracking system is a rich source of movement data, which can help to estimate fatigue or postural instability with no additional cost, or distraction to the user.

Techniques to reduce motion sickness



- Increasing expose time
 - to adapt the experience over time.
- Reducing the display field of view
 - The goal is to reduce the amount of visual information. Then it may mitigate the degree of motion sickness.
- Blur processing
 - The main goal is to reduce the sensitivity of the visual information.

Reducing the display field of view



Dynamically, yet subtly, changing a physically stationary person's FOV in response to visually perceived motion as they virtually traverse a VE.

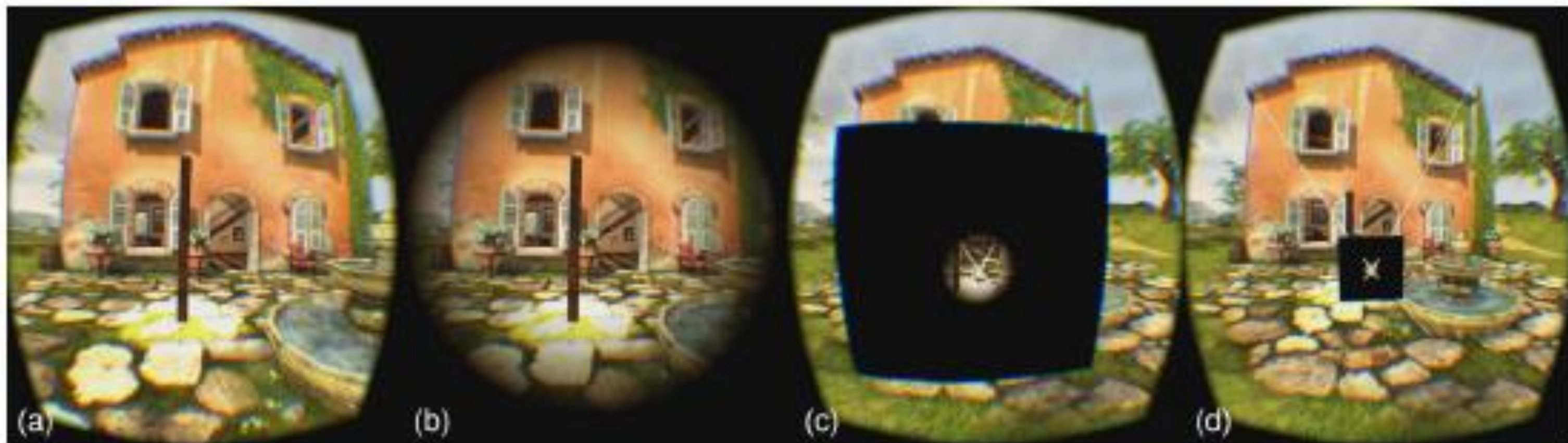
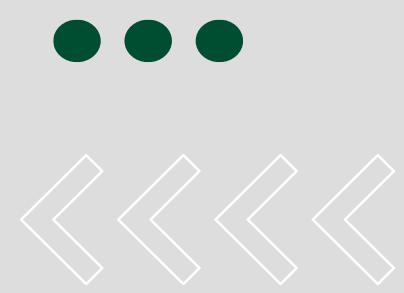


Figure 1: (a) View of VE from one eye with no FOV modification. Study waypoint (lit vertical post) appears at center. (b) Same view restricted by 90° FOV soft-edged circular cutout. While clearly visible at small size shown here, slow restriction to this FOV was imperceptible to majority of study participants when viewed in HWD. (c) Geometry of FOV restrictor and view frustum used to create subfigure a. Cutout is opened wide enough to not affect FOV. (d) Geometry of FOV restrictor and view frustum used to create subfigure b, viewed from same location as in subfigure c. Note that FOV restrictor is scaled about its center in its plane and does not translate relative to center of projection.

Video: <https://www.youtube.com/watch?v=IHzCmfuJYa4>

Mask-based image processing



Reducing motion sickness for rotating movements in VR.

Mask size

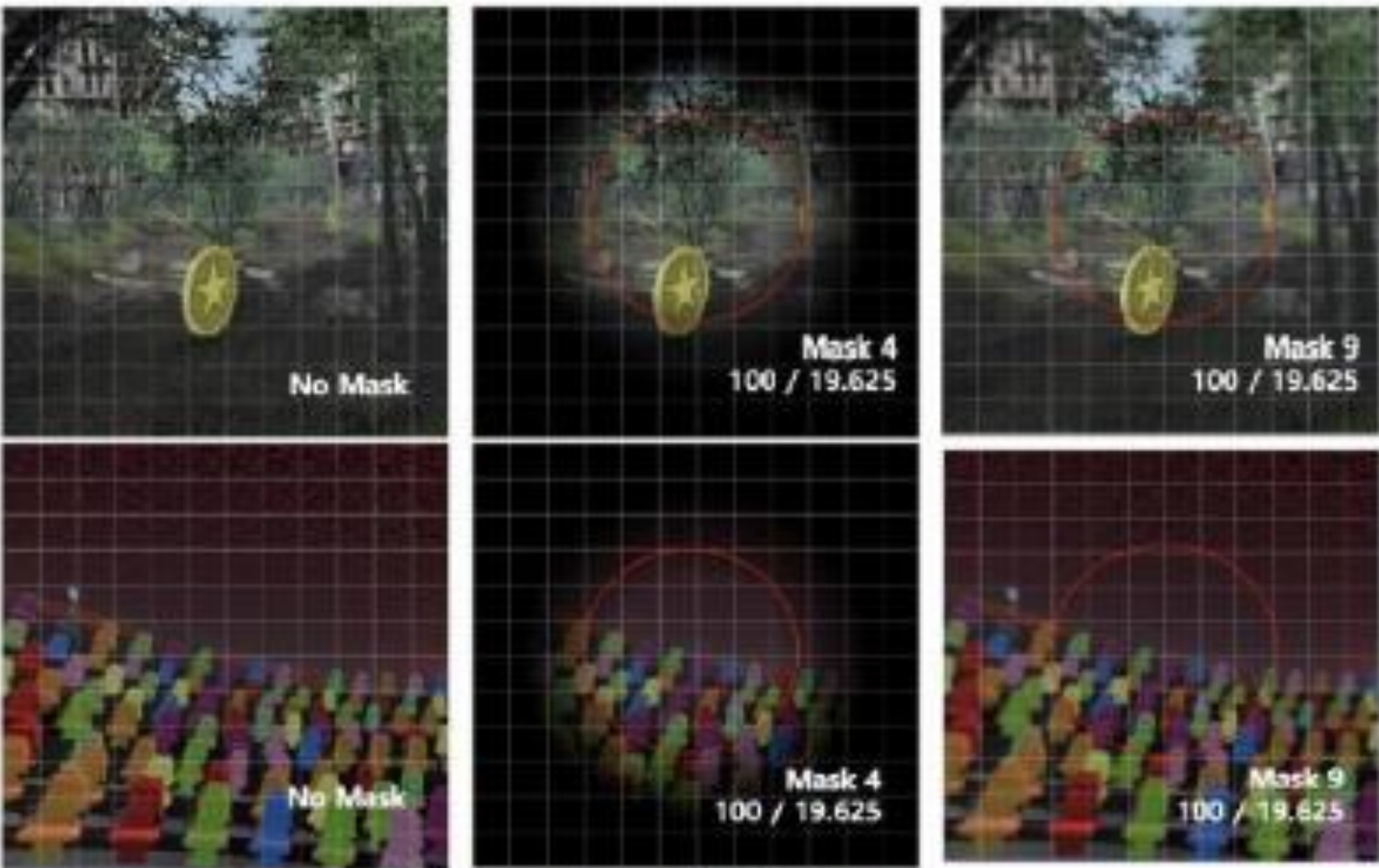


Figure 1. 'Combating VR Sickness through Subtle Dynamic

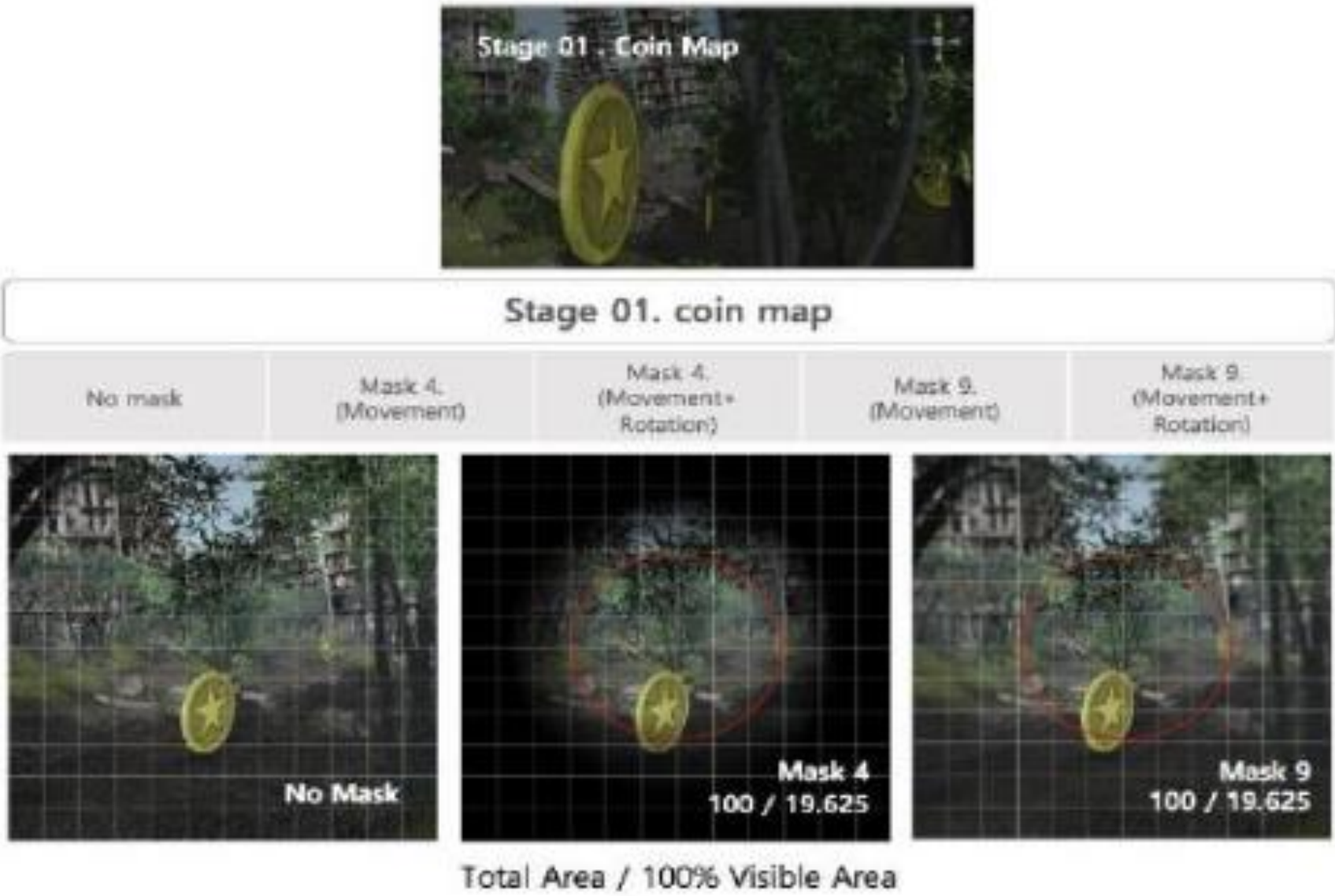
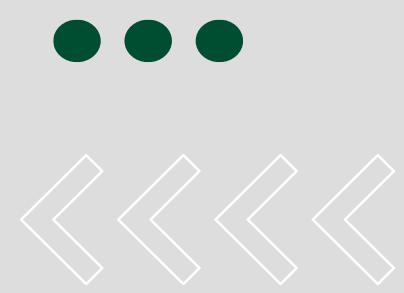


Figure 2. Coin Map (Stage 1)

Mask-based image processing



Reducing motion sickness for rotating movements in VR.

Please answer only in pre-experiment.				
		No Mask (1)	Black circular (2)	Black square (3)
5	List the mask in the order of lowest motion sickness. Ex) Mask with the least motion sickness : 1, Mask with the more motion sickness : 9			
		Black gradation circular (4)	Black gradation circular (5)	Gray gradation circular (6)
		White gradation circular_100 (7)	White gradation circular_50 (8)	Gaussian blur (9)

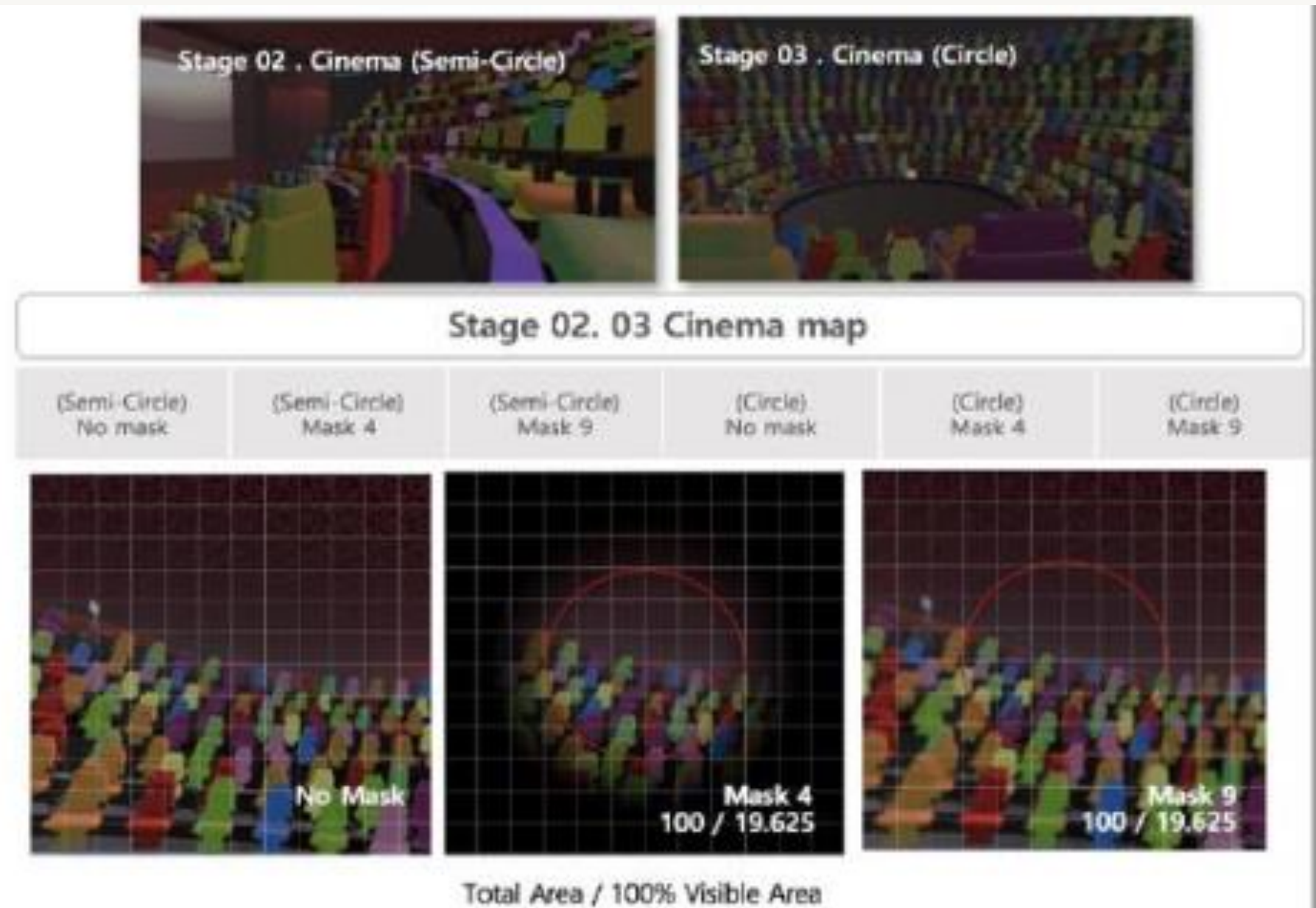


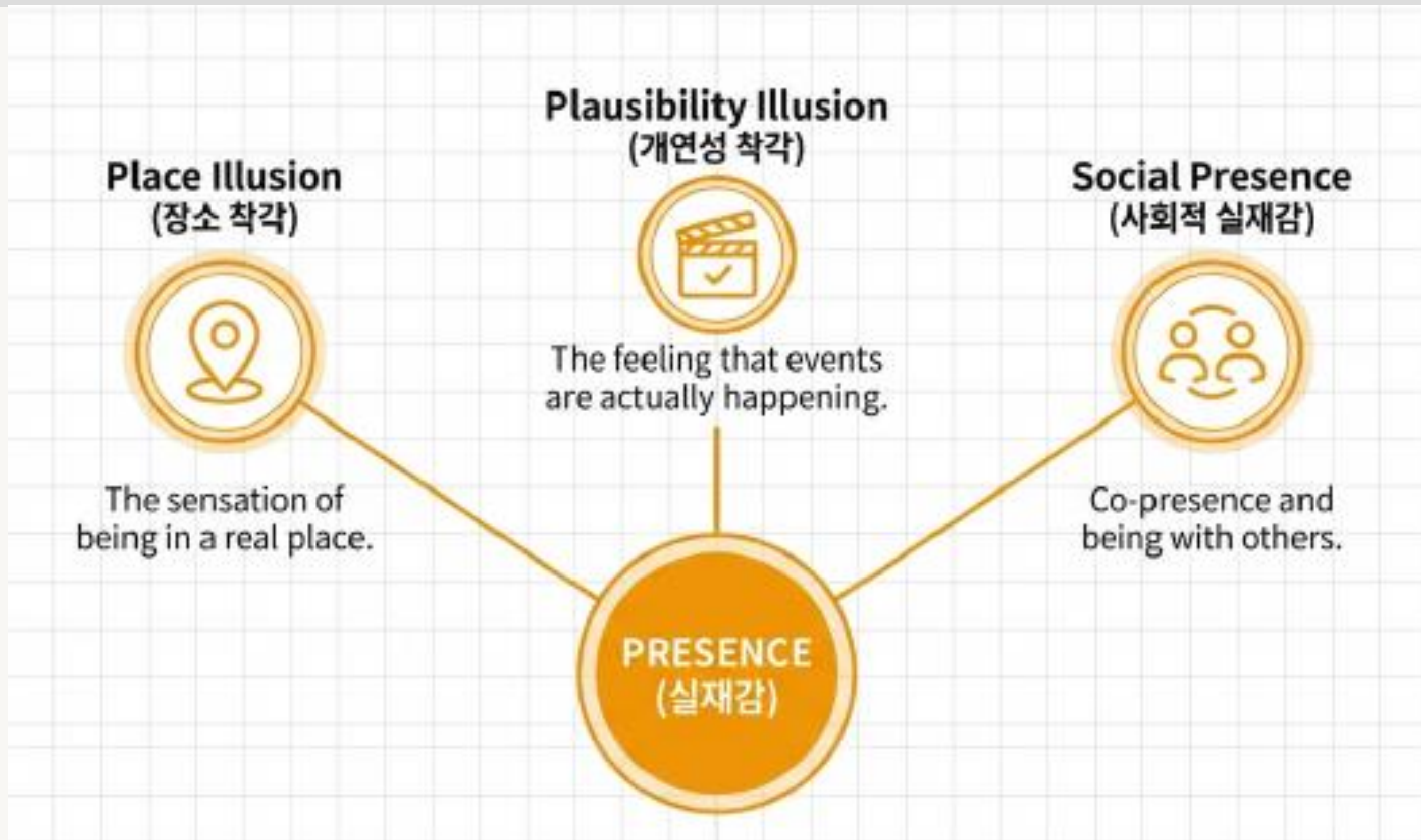
Figure 3. Cinema Map (Stage 2, 3)



PRESENCE



What is presence?



What is presence?



The most common definition is “Being there”.

- Transportation to another place
- Place Illusion
- Co-presence
- Spatial presence
- Social presence

Measuring Presence



Four categories

- Self-report (questionnaires),
- Behavioral (response to threatening stimuli),
- Physiological (electrodermal activity),
- and psychophysical measures (subjective comparison evaluation)

Self-report

- This is a main evaluation method for presence.
- Self-report refers to all techniques in which users actively report some information about their experience to the experimenter.
 - Post-experience questionnaires

Table 3. List of Presence/Telepresence Questionnaires

Questionnaire	# Items	Subscales	Intended use
Slater-Usuh-Steed (SUS) questionnaire [147]	6	No separate subscales	Virtual environments
Arrival/Departure [71]	8	Arrival; Departure	Cross-media
Witmer-Singer Presence Questionnaire (PQ) [157]	19	Involved/Control; Natural; Interface Quality	Virtual environments
Lombard et al. questionnaire [81]	103	Social richness; Realism; Transportation; Immersion; Social actor within a medium; Medium is a social actor	Cross-media
Reality Judgment and Presence Questionnaire (RJPQ) [8]	77	Reality judgment; Presence; Emotional involvement; Interaction; control; Attention/Flow; Realism, Congruence/Continuity; Expectations	Virtual environments
Swedish Viewer-User Presence (SVUP) questionnaire [75]	150	"quality evaluations, attitudes, presence, and realism, and information from different modalities as well as simulation sickness items"	Virtual environments
ITC-Sense of Presence Inventory (ITC-SOPI) [78]	44	Sense of physical space; Engagement; Naturalness; Negative effects	Cross-media
Igroup Presence Questionnaire (IPQ) [109]	14	Presence; Spatial presence; Involvement; Realness	Virtual environments
Sas and O'Hare questionnaire [105]	34	Being there; Not being here; Reflective consciousness	Virtual environments
MEC-Spatial Presence Questionnaire (MEC-SPQ) [149]	L: 72; M: 54; S: 36	Attention allocation; Spatial situation model; Self-location; Possible actions; Cognitive involvement; Suspension of disbelief; Domain-specific interest; Visual/spatial imagery; absorption	Cross-media
Bouchard et al. questionnaire [24]	1	No separate subscales	Virtual environments
Experimental Virtual Environment-Experience Questionnaire (EVEQ) [141]	124	Physical presence; Situational involvement; Competence	Virtual environments
Temple Presence Inventory (TPI) [82]	42	Transportation; Immersion; Realism; Social actor within a medium; Social richness	Cross-media
Virtual Experience Test (VET) [36]	17	Affective; Cognitive; Sensory; Active; Relational	Cross-media

limitations of self-reporting



- 표준 질문지 항목 (Standard Questionnaire Items)
 1. 당신이 그곳에 있다는 느낌을 1-7로 평가하십시오
(Rate your sense of being there 1-7)
 2. 가상 세계가 더 현실적으로 느껴졌습니까?
(Did the virtual world seem more real?)
 3. 가상 환경에서 얼마나 몰입감을 느꼈습니까?
(How immersive did you feel in the virtual environment?)
 4. 가상 물체와 상호작용하는 것이 자연스러웠습니까?
(Was interacting with virtual objects natural?)

The Paradox (역설)

사용자에게 ‘당신은 거기에 있습니까?’라고 묻는 순간, 그들은 자신이 거기에 없다는 것을 즉시 상기하게 됩니다.

자기 보고는 침해적이며, 측정하고자 하는 환상 자체를 깨뜨립니다.



Slater-Usoh-Steed (SUS) Questionnaires for VEs

1. Please rate your sense of being in the virtual environment, on a scale of 1 to 7, where 7 represents your normal experience of being in a place.
2. To what extent were there times during the experience when the virtual environment was the reality for you?
3. When you think back to the experience, do you think of the virtual environment more as images that you saw or more as somewhere that you visited?
4. During the time of the experience, which was the strongest on the whole, your sense of being in the virtual environment or of being elsewhere?
5. Consider your memory of being in the virtual environment. How similar in terms of the structure of the memory is this to the structure of the memory of other places you have been today? By 'structure of the memory' consider things like the extent to which you have a visual memory of the virtual environment, whether that memory is in color, the extent to which the memory seems vivid or realistic, its size, location in your imagination, the extent to which it is panoramic in your imagination, and other such structural elements.
6. During the time of your experience, did you often think to yourself that you were actually in the virtual environment?

Arrival and departure: psychological movement of the sense of reality ●●●



Arrival/Departure Questionnaires for Cross-media



Arrival is the feeling of being there in a mediated environment;

departure is the feeling of not being in the real environment.

FACTOR 1: A factor score for these items was calculated and labeled “arrival” (Being Present in the Mediated Environment).

1. When the broadcast ended, I felt like I came back to the “real world” after a journey.
2. The television came to me and created a new world for me, and the world suddenly disappeared when the broadcast ended.
3. During the broadcast, I felt I was in the world the television created.
4. During the broadcast, my body was in the room, but my mind was inside the world created by television.
5. During the broadcast, the television-generated world was more real or present for me compared to the “real world.”

FACTOR 2: A factor score of these items, reversed, was calculated and labeled “departure” (Not Being Present in the Unmediated Environment).

1. During the broadcast, I NEVER forgot that I was in the middle of an experiment.
2. The television-generated world seemed to me only “something I saw” rather than “somewhere I visited.”
3. During the broadcast, my mind was in the room, not in the world created by television.

* The factor loadings are shown below (Rotation = VARIMAX):

Witmer and Singer's Presence Questionnaires (PQ)



Table 2. Presence Questionnaire Item Stems (Version 2.0)

Item Stems	Factors	Subscale	ITCorr
1. How much were you able to control events?	CF	INV/C	0.43*
2. How responsive was the environment to actions that you initiated (or performed)?	CF	INV/C	0.56*
3. How natural did your interactions with the environment seem?	CF	NATRL	0.61*
4. How completely were <i>all</i> of your senses engaged?	SF		0.39*
5. How much did the visual aspects of the environment involve you?	SF	INV/C	0.48*
6. How much did the auditory aspects of the environment involve you?	SF	AUD ^a	0.32*
7. How natural was the mechanism which controlled movement through the environment?	CF	NATRL	0.62*
8. How aware were you of events occurring in the real world around you?	DF		0.03
9. How aware were you of your display and control devices?	DF		−0.14
10. How compelling was your sense of objects moving through space?	SF	INV/C	0.51*
11. How inconsistent or disconnected was the information coming from your various senses?	RF		0.33*
12. How much did your experiences in the virtual environment seem consistent with your real-world experiences?	RF, CF	NATRL	0.62*
13. Were you able to anticipate what would happen next in response to the actions that you performed?	CF	INV/C	0.43*
14. How completely were you able to actively survey or search the environment using vision?	RF, CF, SF	INV/C	0.59*
15. How well could you identify sounds?	RF, SF	AUD ^a	0.34*
16. How well could you localize sounds?	RF, SF	AUD ^a	0.30*
17. How well could you actively survey or search the virtual environment using touch?	RF, SF	HAPTC ^b	0.15
18. How compelling was your sense of moving around inside the virtual environment?	SF	INV/C	0.62*
19. How closely were you able to examine objects?	SF	RESOL	0.55*
20. How well could you examine objects from multiple viewpoints?	SF	RESOL	0.49*
21. How well could you move or manipulate objects in the virtual environment?	CF	HAPTC ^b	0.11
22. To what degree did you feel confused or disoriented at the beginning of breaks or at the end of the experimental session?	RF		−0.06
23. How involved were you in the virtual environment experience?		INV/C	0.52*
24. How distracting was the control mechanism?	DF		0.37*
25. How much delay did you experience between your actions and expected outcomes?	CF	INV/C	0.41*
26. How quickly did you adjust to the virtual environment experience?	CF	INV/C	0.42*
27. How proficient in moving and interacting with the virtual environment did you feel at the end of the experience?	CF	INV/C	0.46*
28. How much did the visual display quality interfere or distract you from performing assigned tasks or required activities?	DF	IFQUAL	0.44*

Table 2. (Continued)

Item Stems	Factors	Subscale	ITCorr
29. How much did the control devices interfere with the performance of assigned tasks or with other activities?	DF, CF	IFQUAL	0.44*
30. How well could you concentrate on the assigned tasks or required activities rather than on the mechanisms used to perform those tasks or activities?	DF	IFQUAL	0.51*
31. Did you learn new techniques that enabled you to improve your performance?	CF		0.33*
32. Were you involved in the experimental task to the extent that you lost track of time?		INV/C	0.41*

Note. Major Factor Category: CF = Control Factors, SF = Sensory Factors, DF = Distraction Factors, RF = Realism Factors. Subscales: INV/C = Involvement/Control, NAT = Natural, AUD = Auditory, HAPTC = Haptic, RES = Resolution, IFQUAL = Interface Quality. ITCorr = Pearson correlation coefficients between PQ item scores and the PQ Total Score.

^aNo auditory stimulation was provided in our experiments.

^bNo haptic stimulation was provided in our experiments.

* $p < .001$

Immersive Tendency Questionnaires (ITQ)



Table 3. Immersive Tendency Questionnaire Item Stems (Version 2.0)

Item Stems	Subscale	ITCorr
1. Do you ever get extremely involved in projects that are assigned to you by your boss or your instructor, to the exclusion of other tasks?		0.26*
2. How easily can you switch your attention from the task in which you are currently involved to a new task?		0.26*
3. How frequently do you get emotionally involved (angry, sad, or happy) in the news stories that you read or hear?		0.27*
4. How well do you feel today?		0.20
5. Do you easily become deeply involved in movies or TV dramas?	FOCUS	0.49**
6. Do you ever become so involved in a television program or book that people have problems getting your attention?	INVOL	0.47**
7. How mentally alert do you feel at the present time?	FOCUS	0.40**
8. Do you ever become so involved in a movie that you are not aware of things happening around you?	INVOL	0.56**
9. How frequently do you find yourself closely identifying with the characters in a story line?	INVOL	0.53**
10. Do you ever become so involved in a video game that it is as if you are inside the game rather than moving a joystick and watching the screen?	GAMES	0.55**
11. On average, how many books do you read for enjoyment in a month?		0.16
12. What kind of books do you read most frequently? (CIRCLE ONE ITEM ONLY!)		
Spy novels Fantasies Science fiction		
Adventure Romance novels Historical novels		
Westerns Mysteries Other fiction		
Biographies Autobiographies Other non-fiction		
13. How physically fit do you feel today?	FOCUS	0.30**
14. How good are you at blocking out external distractions when you are involved in something?	FOCUS	0.46**
15. When watching sports, do you ever become so involved in the game that you react as if you were one of the players?		0.43**
16. Do you ever become so involved in a daydream that you are not aware of things happening around you?	INVOL	0.56**
17. Do you ever have dreams that are so real that you feel disoriented when you awake?	INVOL	0.50**
18. When playing sports, do you become so involved in the game that you lose track of time?	FOCUS	0.46**
19. Are you easily disturbed when working on a task?		-0.03
20. How well do you concentrate on enjoyable activities?		0.49**
21. How often do you play arcade or video games? (OFTEN should be taken to mean every day or every two days, on average.)	GAMES	0.35**

Table 3 (Continued)

Item Stems	Subscale	ITCorr
22. How well do you concentrate on disagreeable tasks?		0.29**
23. Have you ever gotten excited during a chase or fight scene on TV or in the movies?	FOCUS	0.51**
24. To what extent have you dwelled on personal problems in the last 48 hours?		-0.10
25. Have you ever gotten scared by something happening on a TV show or in a movie?	INVOL	0.42**
26. Have you ever remained apprehensive or fearful long after watching a scary movie?	INVOL	0.31**
27. Do you ever avoid carnival or fairground rides because they are too scary?		-0.05
28. How frequently do you watch TV soap operas or docu-dramas?		0.28**
29. Do you ever become so involved in doing something that you lose all track of time?	FOCUS	0.49**

Note. Subscales: INVOL = Tendency to become involved in activities, FOCUS = Tendency to maintain focus on current activities, GAMES = Tendency to play video games

Note. ITCorr = Pearson correlation coefficients between ITQ item scores and the ITQ Total Score.

* $p < 0.01$

** $p < 0.001$

Six Dimensions of Presence

- Presence as social richness
- Presence as realism
- Presence as transportation
- Presence as immersion
- Presence as social actor within medium
- Presence as medium as social actor

Reality Judgment Presence Questionnaires for VEs



TABLE 1. FACTORIAL ANALYSE		
Factors	Items	Rotated loadings
Factor 1: Reality judgment	18. To what extent did the experience seem real to you?	.819
	11. In your opinion, how was the quality of the images in the virtual world?	.781
	2. To what extent was what you saw in the virtual world similar to reality?	.742
	37. How real did the virtual objects seem to you?	.729
	38. To what extent was what you experienced in the virtual world congruent to other experiences in the real world?	.712
	17. To what extent did you feel you "went into" the virtual world?	.701
	32. To what extent did your interactions with the virtual world seem natural to you, like those in the real world?	.657
	9. To what extent did you feel you "were" physically in the virtual world?	.543
Factor 2: Internal/external correspondence	40. To what extent could you move around the virtual world?	.806
	60. To what extent were the events in the virtual world congruent to your actions?	.735
	36. To what extent could you interact with the virtual world?	.723
	21. To what extent did the virtual world respond to your actions?	.716
	54. To what extent did your actions produce changes in the virtual world?	.591
	56. To what extent were you yourself while experiencing the virtual environment?	.473
Factor 3: Attention/absorption	61. To what extent did you feel it was necessary to devote all your attention to what you were doing in the virtual world?	.749
	30. To what extent did you feel like you "went into" the virtual world and you almost forgot about the world outside?	.712
	48. To what extent did you have to pay excessive attention to what was going on in the virtual world?	.655
	70. To what extent did you forget you were in a room wearing a helmet?	.606

Analysis of Questionnaires



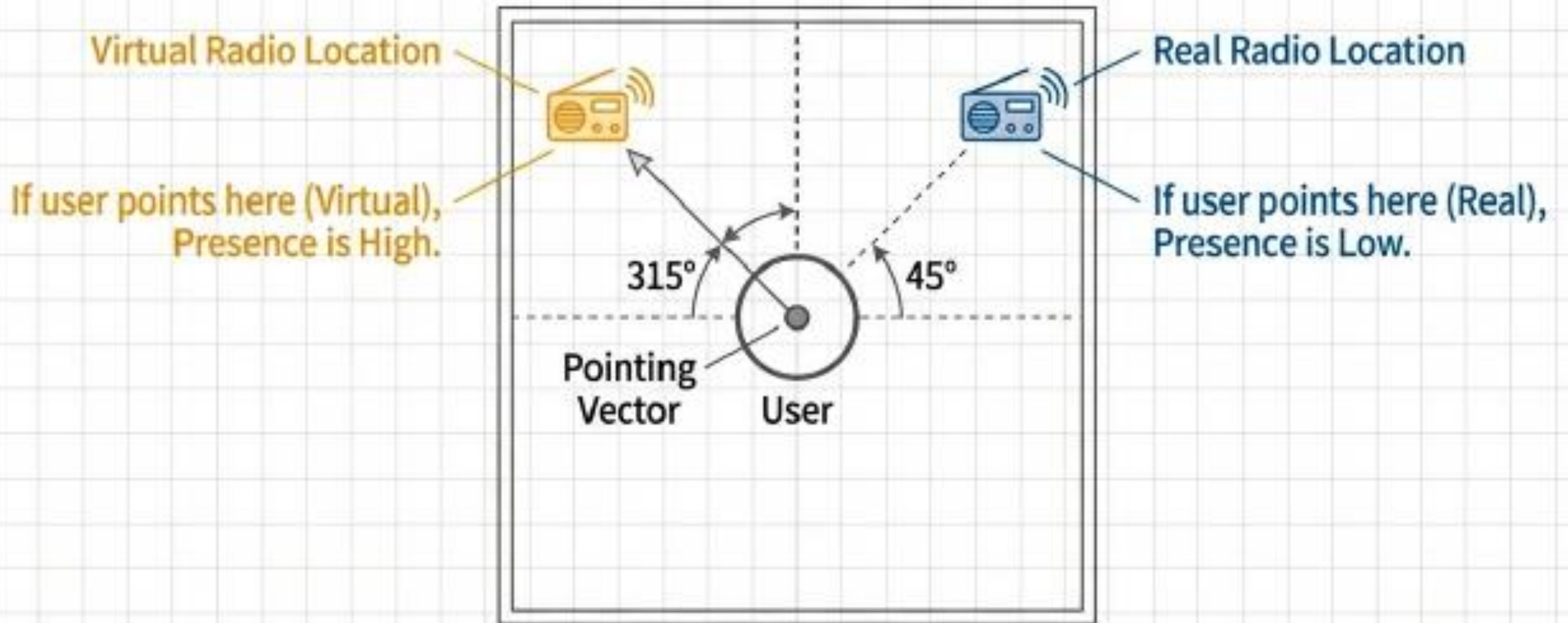
- Convenience
- Valid
- Intrusive but not continuous
- Subjective
- Rely on user's interpretation of potentially difficult concepts to generate meaningful results.

Behavioral Measures



- Presence could be measured by behavioral methods,
- such as response to threatening stimuli (for example, flinching out of the way of a virtual ball),
- or to socially conditioned behaviors (for example, saying, “Gesundheit,” in response to a sneeze in the VE).

Behavioral Measurement: Virtual Radio



Subconscious behavioral pointing reveals true mental location without asking questions.

Task oriented evaluation



Pointing to an ambiguous object as a measure of presence [Slater et al. 1995]

- Participants saw a radio in one location in the real world, and in a different location in the virtual world, and were told at various times during the experiment to point to “the radio.”
- It was assumed that participants who pointed to the virtual radio were experiencing greater presence than participants who pointed to the location of the real radio.
- This was operationalized as a ratio of the differences between the actual pointing angle and the directions to the real and virtual angles.

“Comeback Rate”



- **Participants were told before the experiment to bring something to read, then after the experiment were given the option of reading or re-entering the virtual environment after the experiment had “officially” ended.**
- **They found that users “came back” more often when the experience contained more social presence cues.**

Postural responses to motion stimuli"



- Participants viewed two video clips: one was excerpted from a video recorded from the hood of a rally car, and the other was a still frame from a video taken at the side of the rally track.
- The soundtrack was the same in both videos, giving the impression of a car off in the distance in the still video case.
- All participants saw both stimuli in both monoscopic and stereoscopic presentations.
- Participants rated their presence (as well as involvement, self-motion, and sickness) on a scale from 0 to 100 on a visual analogue scale after each trial, and their postural responses were tracked.

Analysis of behavioral measures



- **Compared to self-report, behavioral measures address one of the major shortcomings, that is, they are objective measures.**
- **Generally contemporaneous**
- **Nonintrusive**
- **Not continuous**

Physiological Measures

- Skin conductance response
- Electrocardiogram (EKG) data
- Skin temperature
- Heart rate

A larger increase in heart rate when exposed to the visual cliff significantly correlated with an increase in presence as measured by the SUS questionnaire.



Figure 3. Subject wearing HMD and physiological monitoring equipment in the "Pit Room".

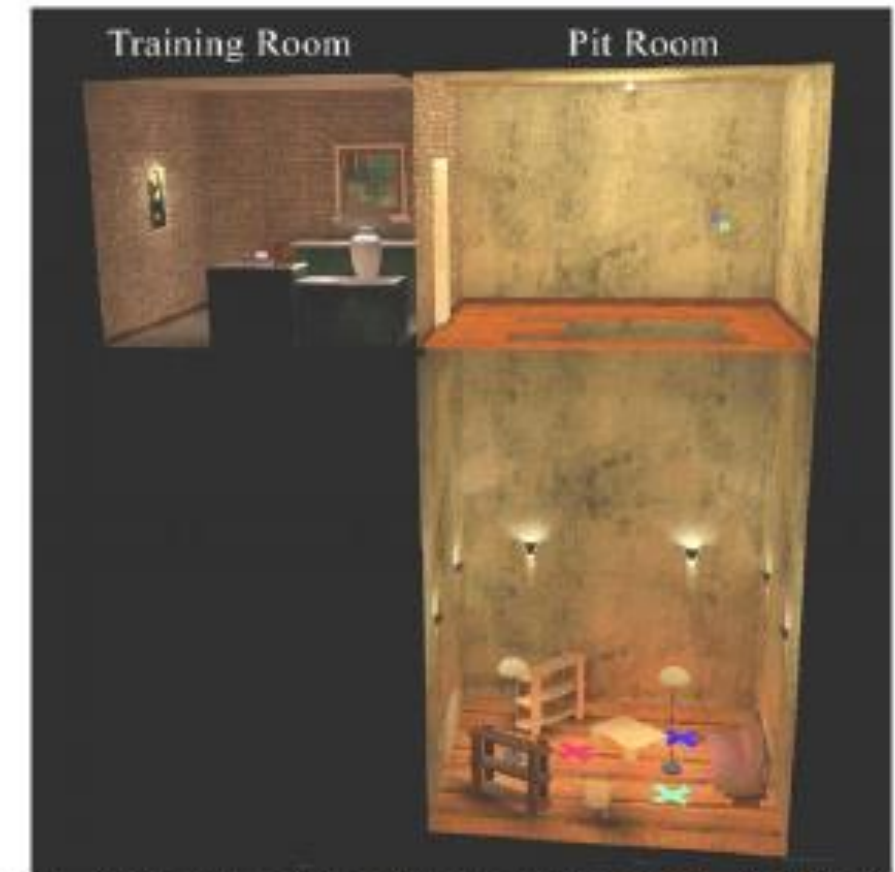
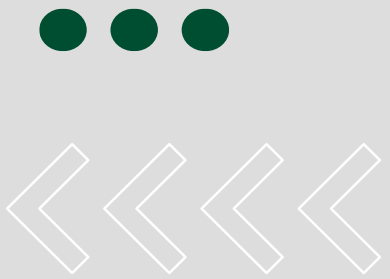


Figure 1. Side view of the virtual environment. Subjects start in the Training Room and later enter the Pit Room.



Figure 2. View of the 20' pit from the wooden ledge.

Physiological Measures

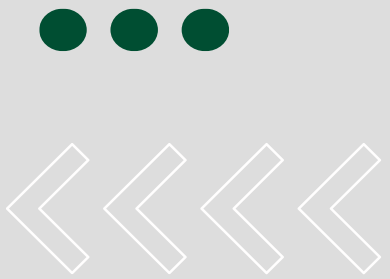


- Understanding the neural region with the sense of presence using fMRI technique [Baumgartner et al. 2008].
- They have identified regions of the brain that seem to correlate with the feeling of presence in participants, the dorsolateral prefrontal cortex, while Bouchard et al. identify the parahippocampus as the brain region most associated with the feeling of presence.
- Baumgartner et al. speculate that this difference may be due to the different study methodologies (Baumgartner et al. presented two different VEs for the two groups in their study, while Bouchard et al. presented the same VE to both groups.)

Thomas Baumgartner, Dominique Speck, Denise Wettstein, Ornella Masnari, Gian Beeli, and Lutz Jäncke. Feeling present in arousing virtual reality worlds: Prefrontal brain regions differentially orchestrate presence experience in adults and children. *Front. Hum. Neurosci.* 2 (2008), 8.

Stéphane Bouchard, Jeanne Talbot, André-Anne Ledoux, Jennifer Phillips, Matteo Cantamesse, and Geneviève Robillard. 2010. Presence is just an illusion: Using fMRI to locate the brain area responsible to the meaning given to places. *Annu. Rev. Cyberther. Telemed.: Adv. Technol. Behav. Soc. Neurosci.* 154 (2010), 193–196.

Analysis of physiological measures



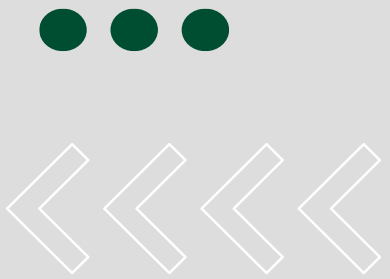
- **Advantages:**

- Physiological measures provide the truly objective evaluation to address one of the major shortcomings of the self-report measures.
- They are contemporaneous and continuous.

- **Disadvantages:**

- Inconvenient and intrusive.
- Not generalizable.
- Additional equipment required.

Psychophysical Methods

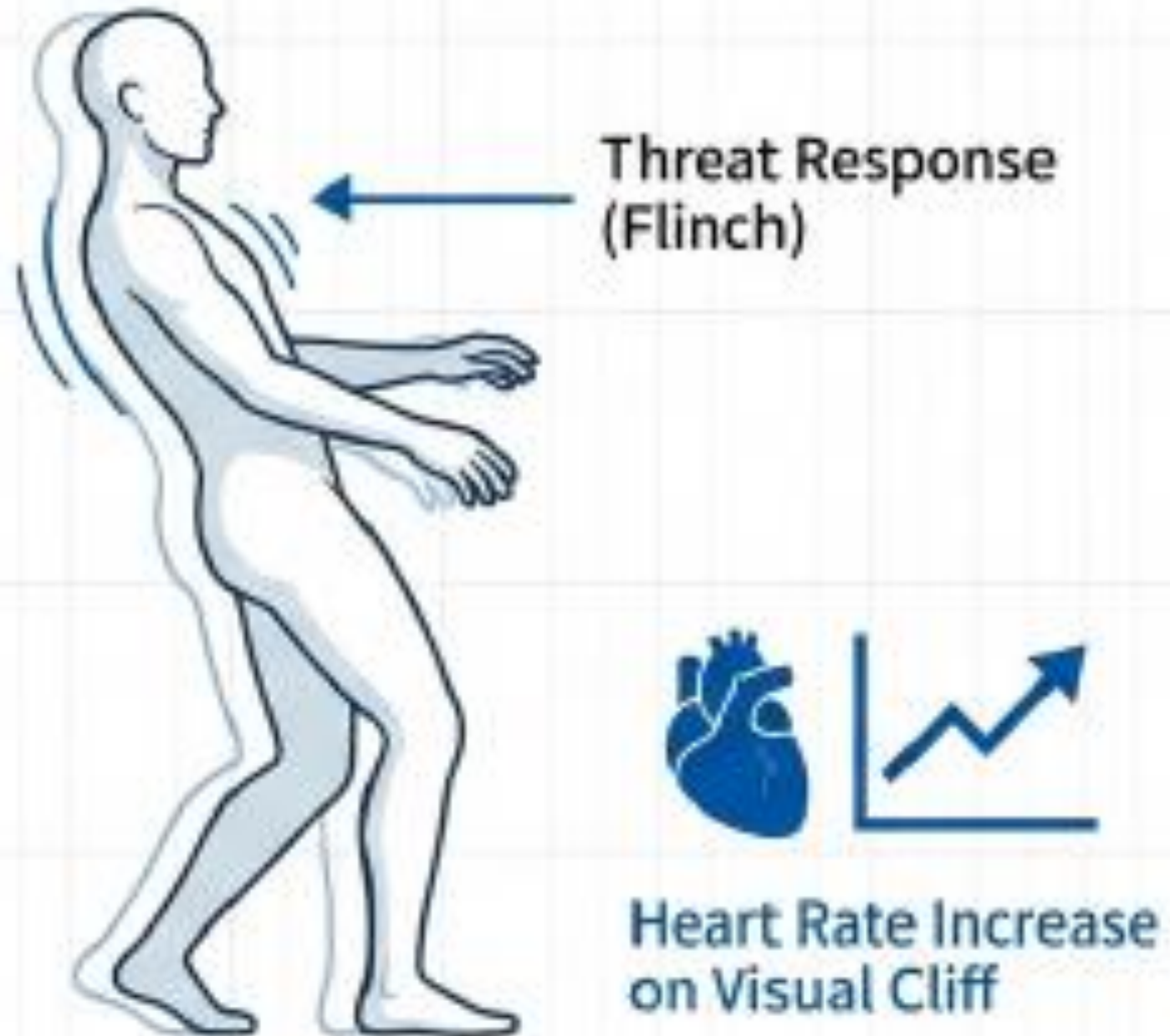


- **An evaluation method of participants' feelings of Place Illusion and Plausibility Illusion.**
- **This method involved**
 - exposing each participant to a VE that was configured with the “highest levels” of various immersion and coherence factors,
 - instructing the participant to remember how much PI or Psi he or she felt in that environment,
 - then exposing the participant to various diminished versions of the virtual environment.

Reaction of the body to the brain

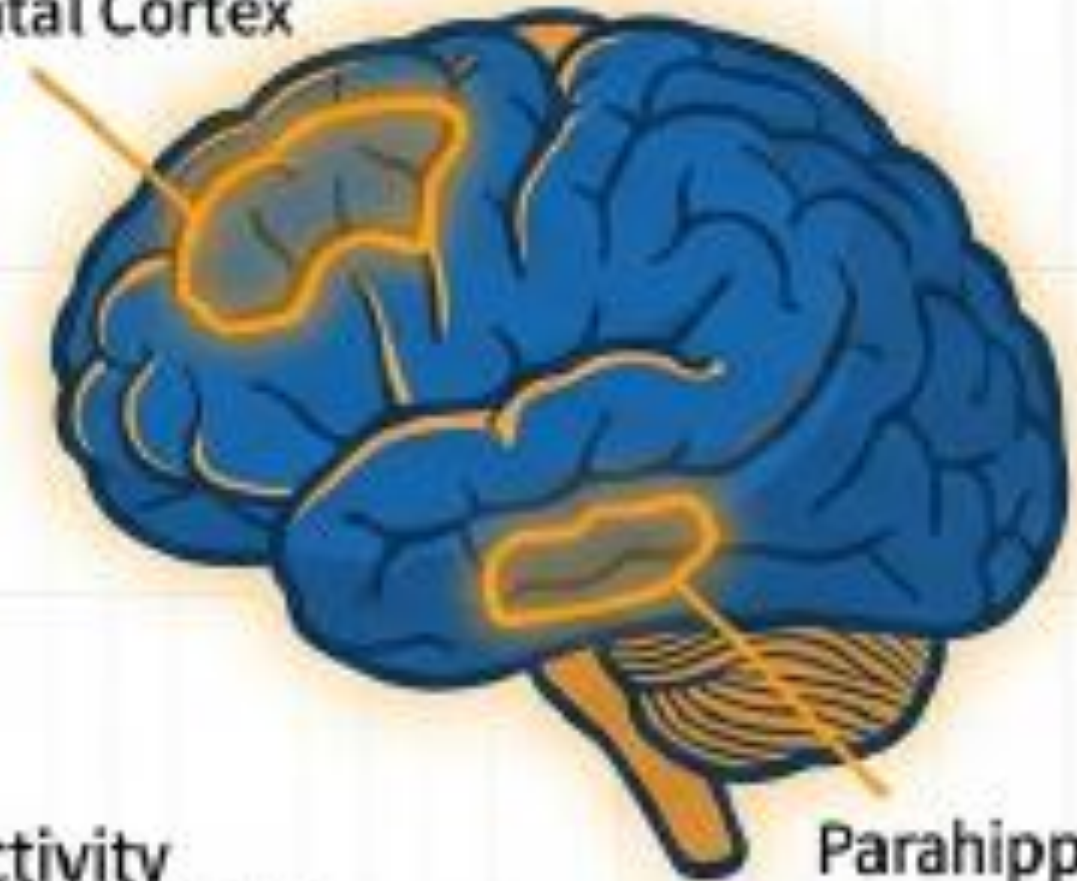


Physiological Reflex (신체 반사)



Neurological Evidence (뇌과학적 증거)

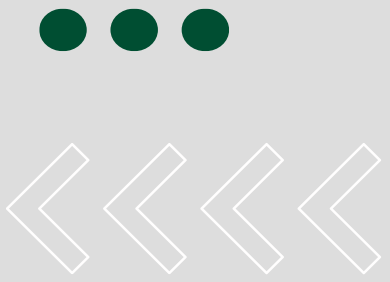
Dorsolateral Prefrontal Cortex



fMRI activity correlates with the sensation of "Being There".

Parahippocampus

Analysis of psychophysical methods



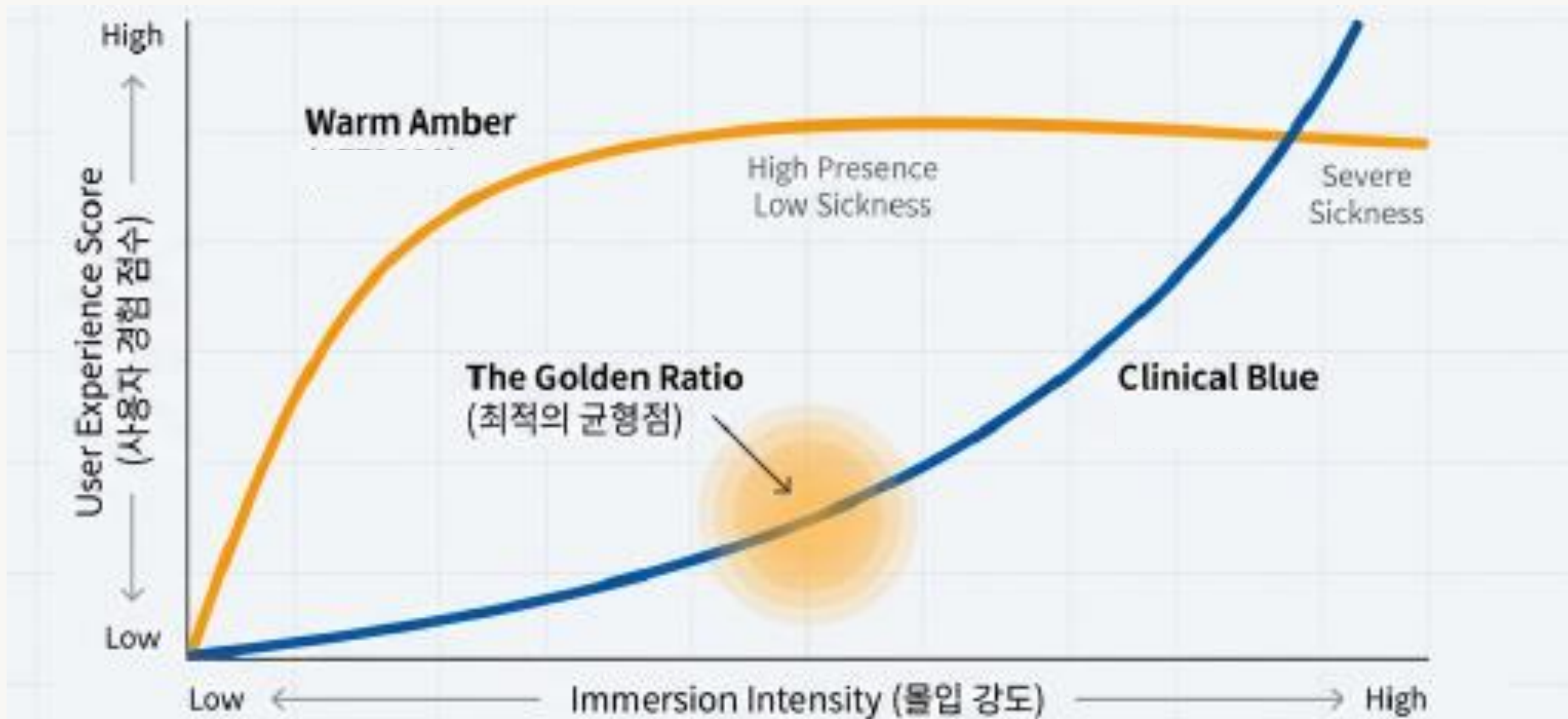
- **Notably unlike all the other mentioned measures.**
- **This evaluation method is to understand how presence is impacted by changes to the immersion or coherence of a virtual experience, other than a degree of presence.**
- **The advantages of this method is no need to conduct extensive tasks, then make the experiment as simple as possible.**

Conclusion



- **This course introduced the measurements of motion sickness and presence for virtual environments.**
- **The advantages and disadvantages of each type of measurement were also briefly summarized.**
- **It is important note that balancing the motion sickness and the sense of presence could be crucial when designing VR applications.**

Golden rain of motion sickness and realism ...



Designing for VR is an optimization problem. Success lies in balancing the minimization of sickness with the maximization of presence.



Sickness in VR

HyungSeok Kim
Konkuk University



Outcomes of Immersion: Three Key Risk Factors

Analysis of key physiological and physical barriers impeding virtual reality experiences



Motion Sickness (멀미)

뇌의 감각 혼란과 생존 본능.



Physical Fatigue (신체 피로)

하드웨어 무게와 부자연스러운 인터랙션.



Hygiene & Safety (위생 및 안전)

감염 위험과 신경학적 부작용.

Types and causes of motion sickness



Motion Sickness (광의의 멀미)



실제 혹은 가상의 움직임에 노출되었을 때 발생하는 부정적 증상

Cybersickness (사이버 멀미)



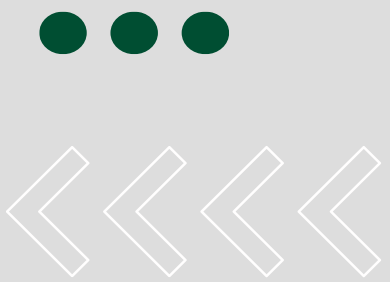
VR 특화 (VR Specific).
물리적 움직임 없이 시각적 자극만으로 발생하는 멀미

Simulator Sickness (시뮬레이터 멀미)



하드웨어 결함 (Hardware Flaws).
콘텐츠가 아닌 시뮬레이션의 기술적 결함(지연, 왜곡)으로 인해 발생

Sensory Conflict Theory: Eye-Ear Discord



결과 (Result)

뇌는 상충되는 정보를 처리하지 못하고 혼란에 빠짐.

뇌의 입장에서 VR은 현실과 다른 '오류' 상태입니다.

Toxin Theory: Vomiting Is a Survival Instinct



독소 이론: 구토는 생존 본능이다

Step 1. 뇌의 해석 (Brain's Interpretation)

 감각 정보의 불일치를
'신경 독소(Neurotoxin)
섭취에 의한 환각'으로 오인.



Step 2. 신체 반응 (Body's Response)

 독소를 배출하기 위해
발한(Sweating)과
구토(Vomiting)를 유도.

진화적 관점: 멀미는 섭취한 독을 뱉어내려는
진화된 방어 기제입니다.

Visual triggers: Movement of scene

The visual movement of the entire masquerade environment, which does not occur in the real world, causes motion sickness.



의도된 움직임 (Intentional Scene Motion)



사용자가 고개를 돌릴 때 세상이 움직임 (정상).



의도하지 않은 움직임 (Unintentional Scene Motion)

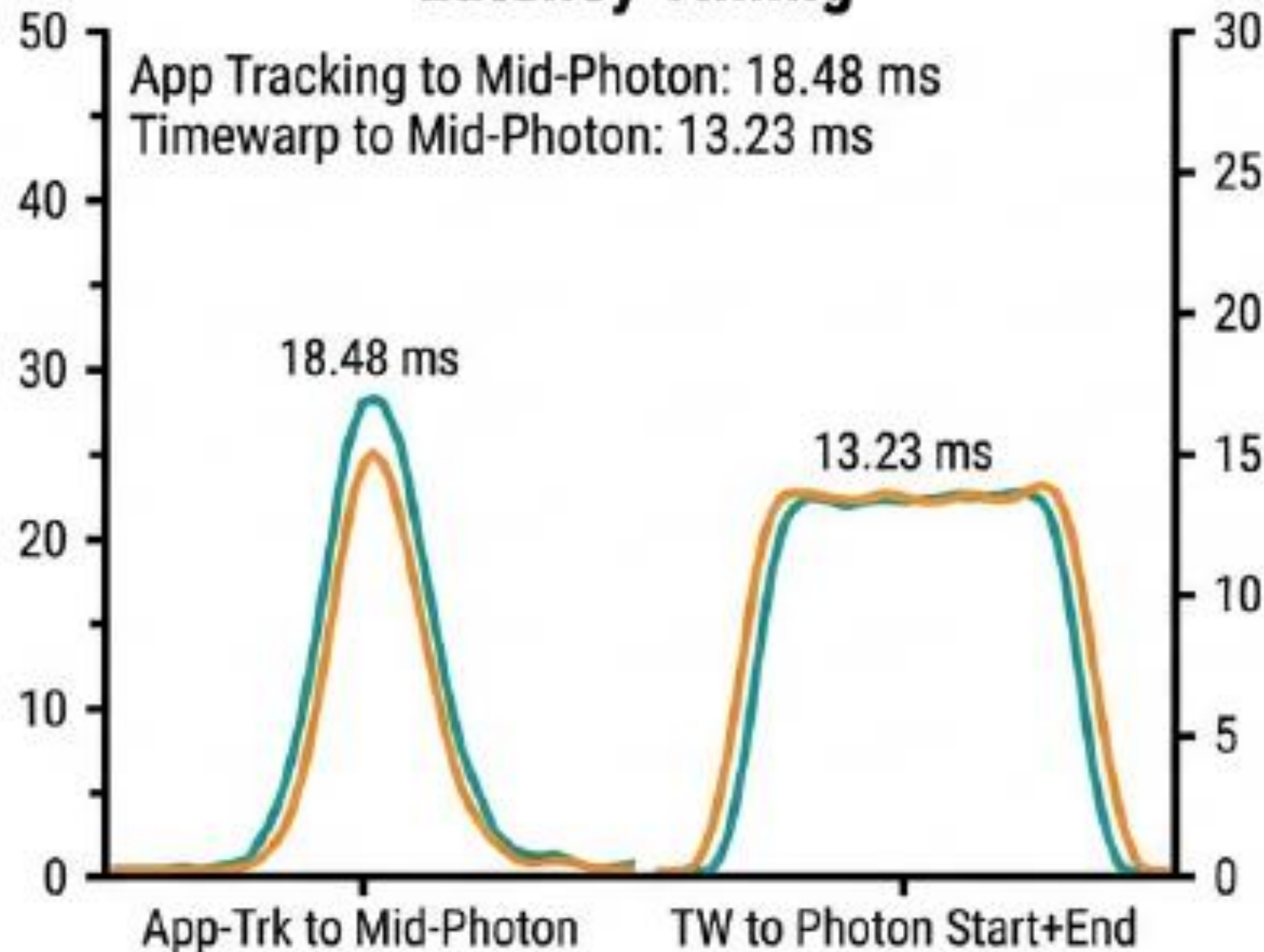


사용자의 입력 없이 카메라가 줌인/줌아웃 하거나 흔들림 (멀미 유발).

Technical fault: Latency and rest frame



Latency Timing



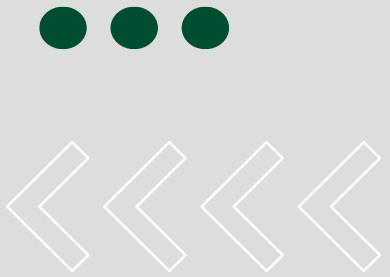
Latency:

Fine delays between behavior and visual responses (e.g., 20ms) add to brain confusion.

Solution: Rest Frame

It should provide a "Stationary Frame of Reference" for the brain to use as a reference (e.g., cockpit)

VR Sickness

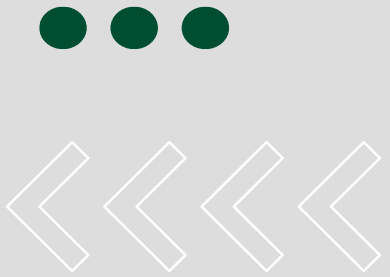


- **Motion Sickness**

- Motion sickness refers to adverse symptoms and readily observable signs that are associated with exposure to real (physical or visual) and/ or apparent motion (Lawson, 2014)

- **Travel Sickness**

Motion Sickness



- **Cybersickness**

Cybersickness is visually induced motion sickness resulting from immersion in a computer-generated virtual world. Motion sickness in VR.

- **Simulator sickness**

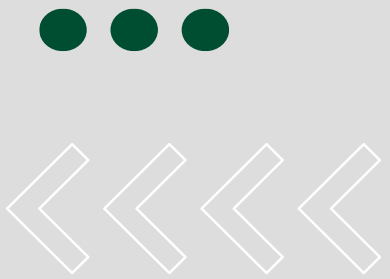
Simulator sickness is sickness that results from shortcomings of the simulation, but not from the actual situation that is being simulated (Pausch et al., 1992).

Motion Sickness



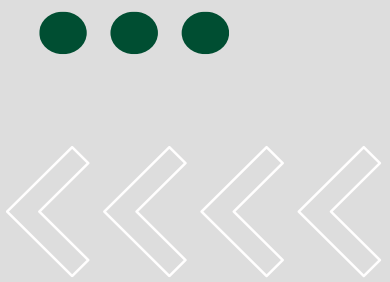
- Motion sickness is a condition in which *a disagreement exists between visually perceived movement and the vestibular system's sense of movement.*
 - Visually induced motion sickness
 - Physically induced motion sickness
- Symptoms of such sickness include general discomfort, nausea, dizziness, headaches, disorientation, vertigo, drowsiness, pallor, sweating, and, in the occasional worst case, vomiting.

Scene Motion in Motion Sickness

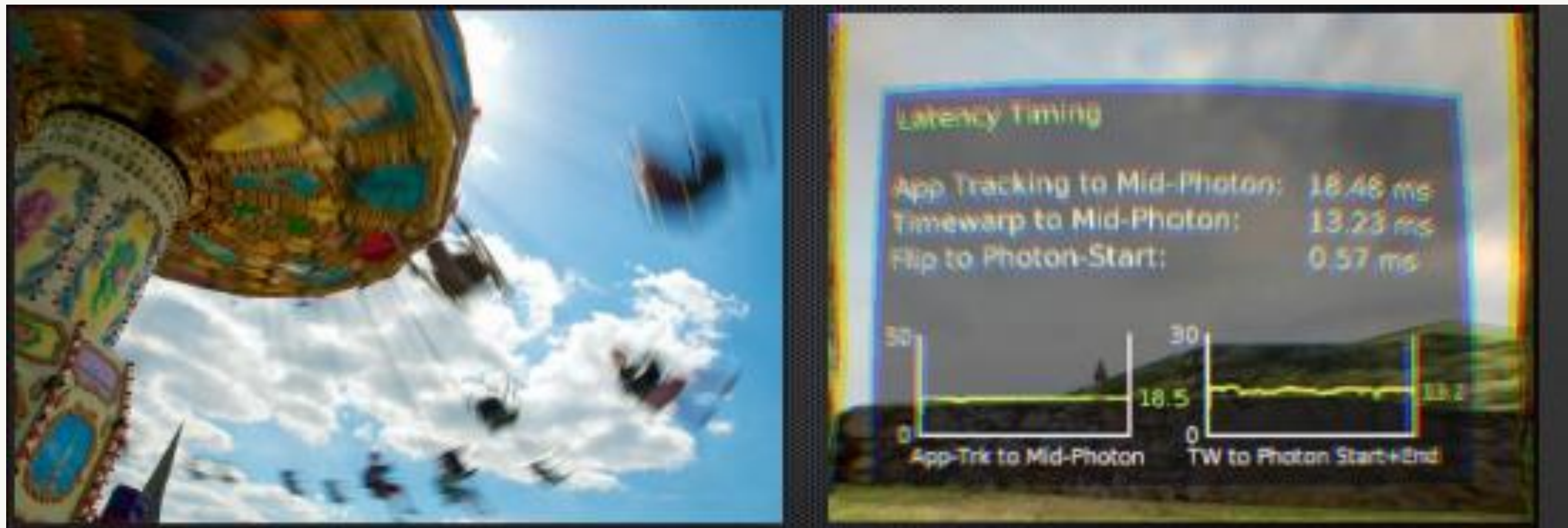


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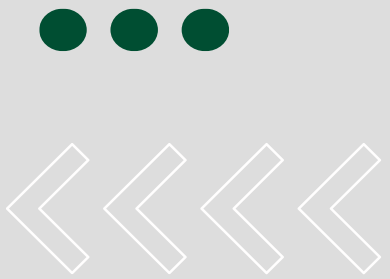
Scene Motion in Motion Sickness



- Scene motion is visual motion of the entire virtual environment that would NOT normally occur in the real world [Jerald2009]
- Intentional scene motion
- Unintentional scene motion



Vection and Motion Sickness



- **Vection**
 - Illusion of self-motion when one is not physically moving in the perceived manner.
- **It doesn't necessarily always cause motion sickness**
 - Why?

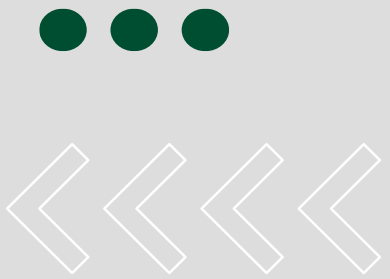
Vection and Motion Sickness



When a visual scene moves independently of how a user is physically moving, there can be mismatch between what is seen and what is physically felt.

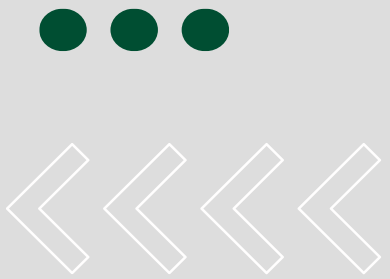


Vection and Motion Sickness



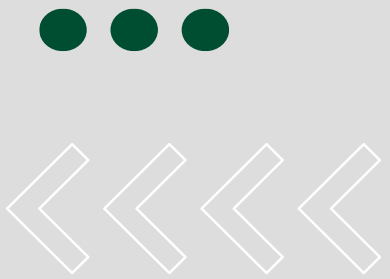
- **Motion sickness is not severe for linear motion**
- **This mismatch is especially discontingenting**
 - when the virtual motion accelerates
 - because the otolith organs do not sense that the same acceleration
- **Virtual rotations can be more discontingenting because the semicircular canals detect both velocity and acceleration [Howard 1986b]—one is more likely to experience motion sickness from constant angular velocity than from constant linear velocity**

Vection and Motion Sickness



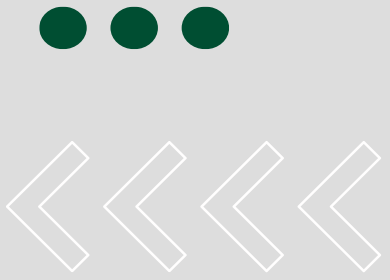
- **Virtual rotations can be more disconting**
 - Because the semicircular canals detect both velocity and acceleration [Howard 1986b]
 - One is more likely to experience motion sickness from constant angular velocity than from constant linear velocity

Theories of Motion Sickness



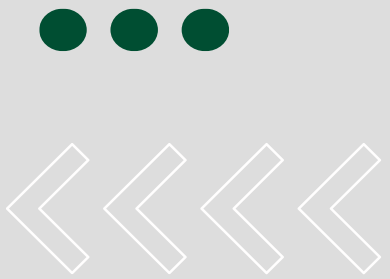
- **Sensory Conflict Theory**
- **Motion sickness may result**
 - when Incoming information (mainly: visual and vestibular) is
 - NOT compatible with each other
 - AND don't match mental modal of expectations
- **In VR**
 - Visual and auditory cues comes from simulation
 - Vestibular and proprioception comes from real!

Theories of Motion Sickness



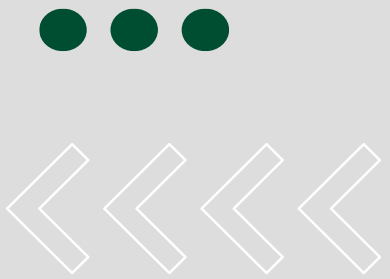
- **Sensory Conflict Theory**
- **One of the great challenges of VR is**
 - To create visual and auditor cues that are consistent
 - With vestibular and proprioceptive cues
- **Sensory Conflict Theory is good to explain most cases**
- **BUT why it happens?**

Evolution Theory



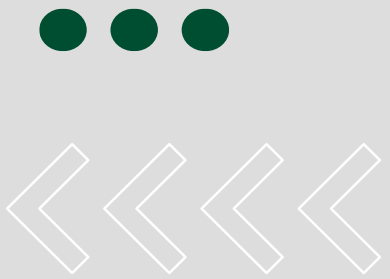
- **The evolutionary theory [Treisman 1977], also known as the poison theory, offers a reason for why motion makes us sick:**
 - It is critical to our survival to properly perceive our body's motion and the motion of the world around us
- **When something is wrong: we get conflict information due to**
 - Toxins, etc
- **Provided with negative actions**
 - Discouraging further actions, sweating and vomiting (to ejecting the poison), and nausea and malaise

Postural Instability Theory



- **When exactly and how severe the sickness will be?**
- **Predicts that sickness results**
 - When an animal lacks or has not yet learned strategies for maintaining postural stability [Riccio and Stoffregen 1991].
- **Postural instability precedes motion sickness (it is the CAUSE of the sickness)**
- **Sickness can be reduced when the user learns how to stabilize posture**

Rest-frame Hypothesis

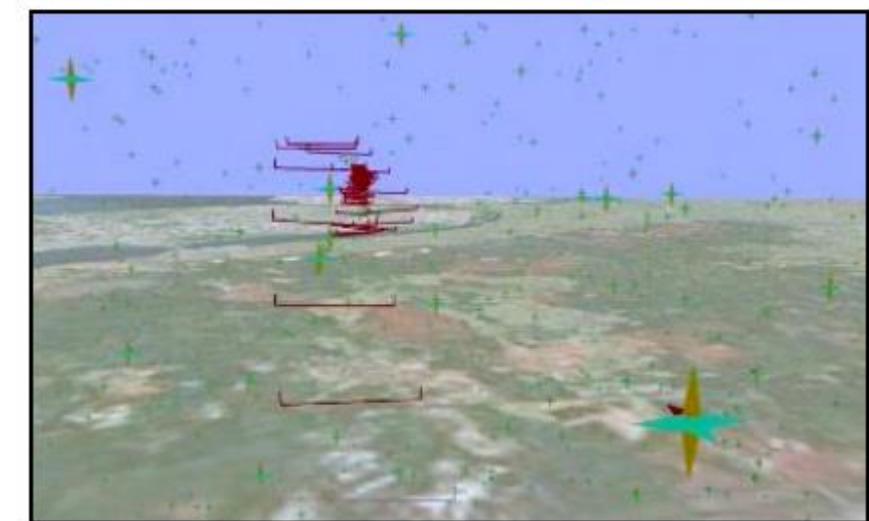
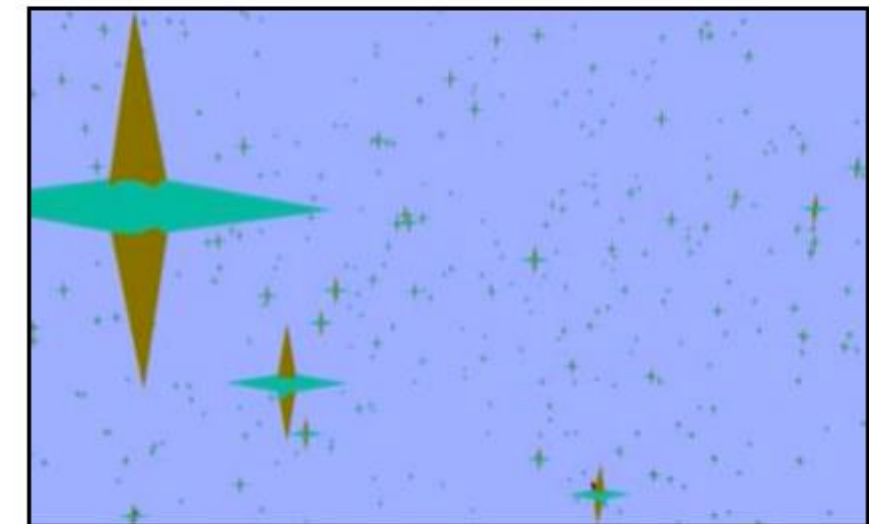
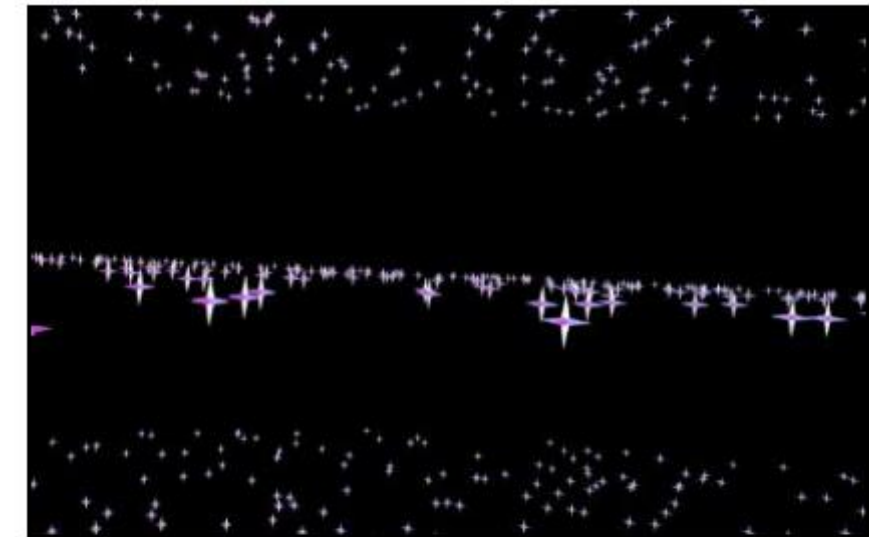


- **Not every sensory conflict induces motion sickness**
- **The rest frame hypothesis**
 - Motion sickness does not arise from conflicting orientation and motion cues directly
 - But rather from conflicting stationary frames of reference implied by those cues [Prothero and Parker 2003]
- **It is necessary to create “rest frame” carefully**

Rest-frame Hypothesis



- **Motion Sickness can be reduced with**
 - proper information on the rest-frame
 - e.g.) 2 DOF vs. 6 DOF
- **Can be reduced by**
 - Providing another rest-frame

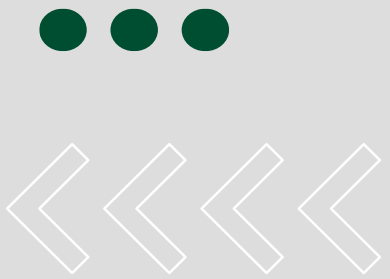


Eye movement Theory



- **Motion sickness occurs due to**
 - The unnatural eye motion required to keep the scene's image stable on the retina.
- **Providing fixation point may help to reduce the motion sickness**

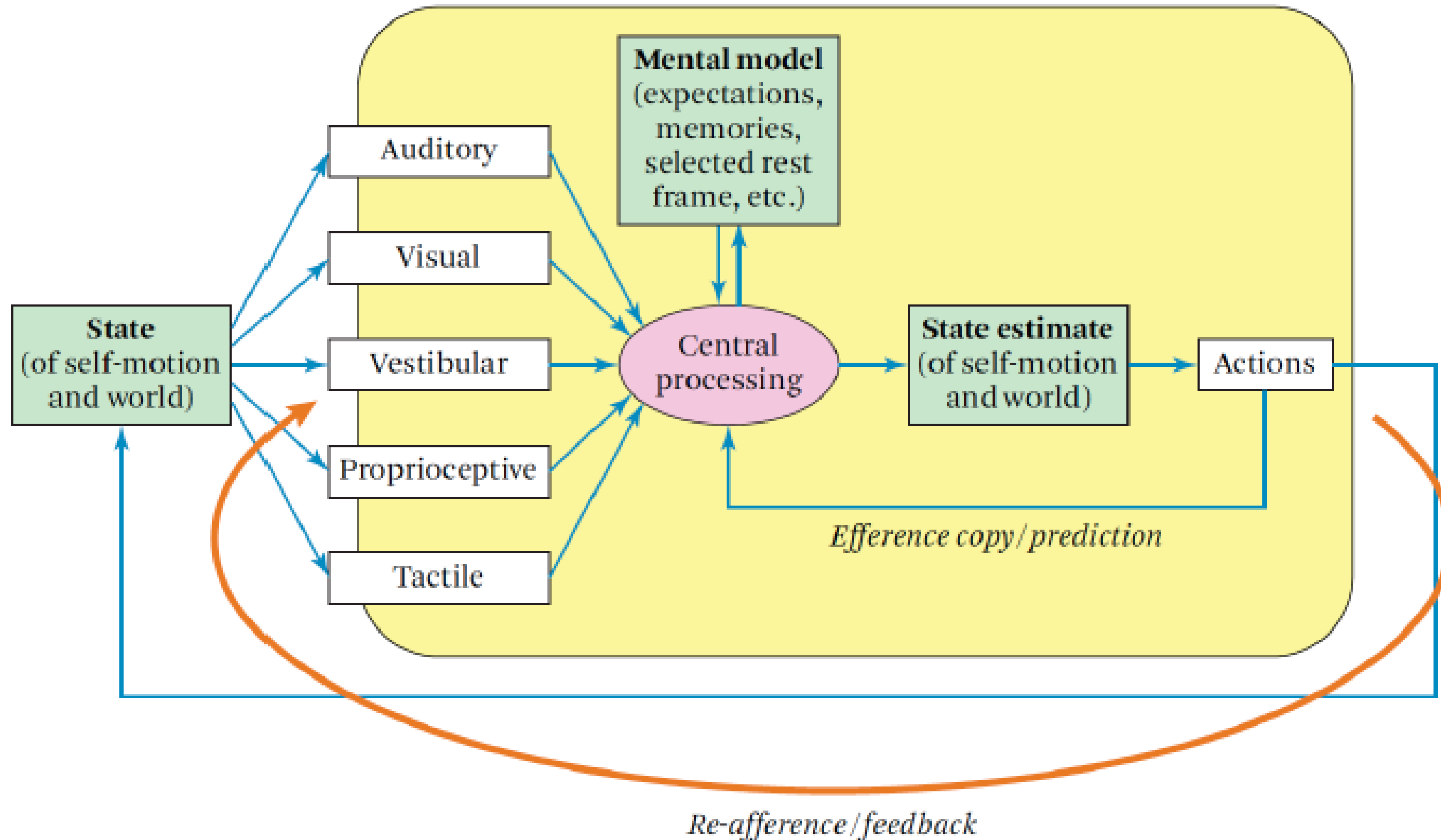
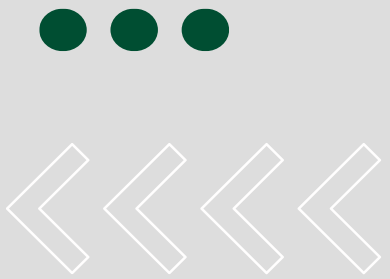
Theories of Motion Sickness



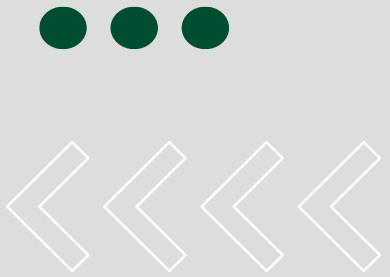
- **Unified model of motion sickness**

The model is consistent with all the previously described theories of motion sickness and can be used to help understand how we perceive motion, whether that motion is perceived as external motion of the world or self-motion, and why motion sickness results

Theories of Motion Sickness



Unified Model



- **Actions include;**
 - Stabilizing actions by adjusting postures
 - Eye movement to stabilize the image
 - Physiological reactions to intoxicating body
- **Motion sickness occurs**
 - When new mental model is created

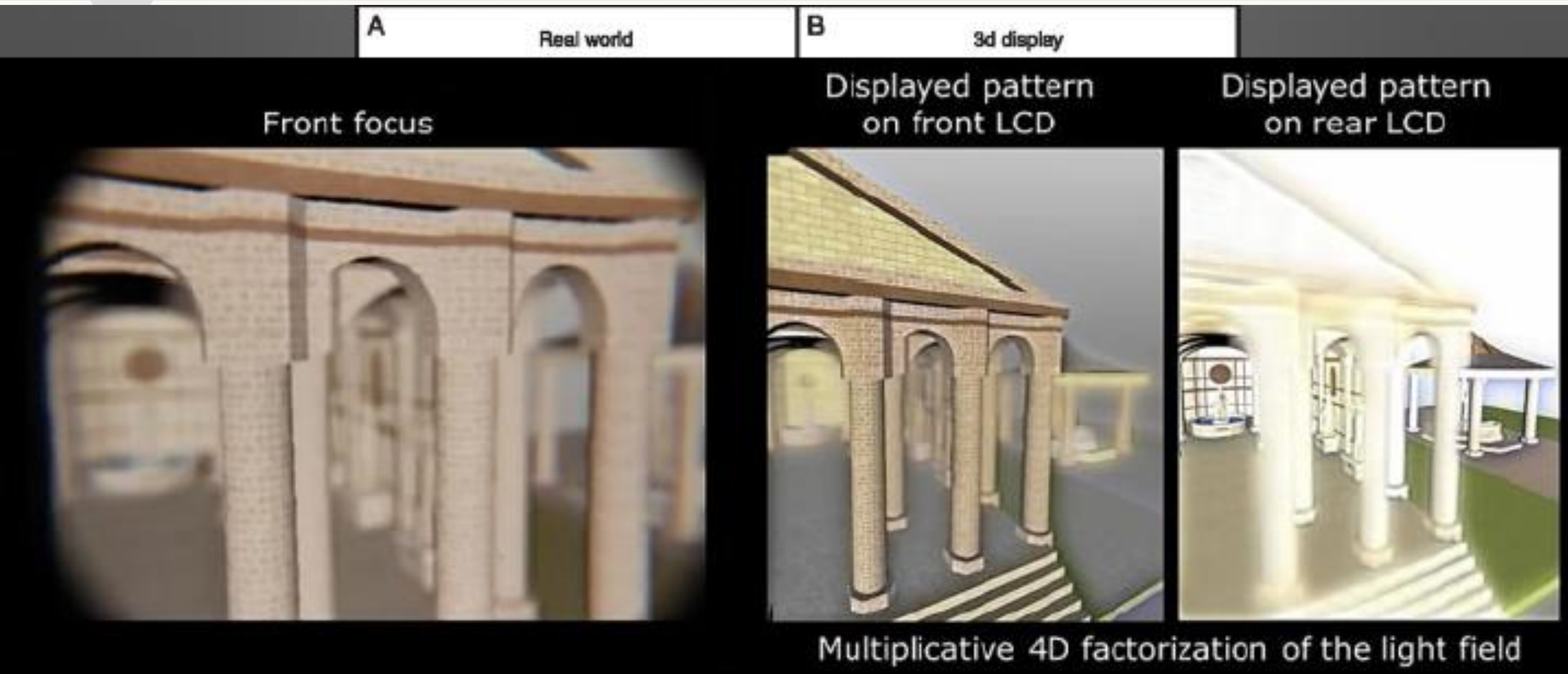
Health Issues of VR



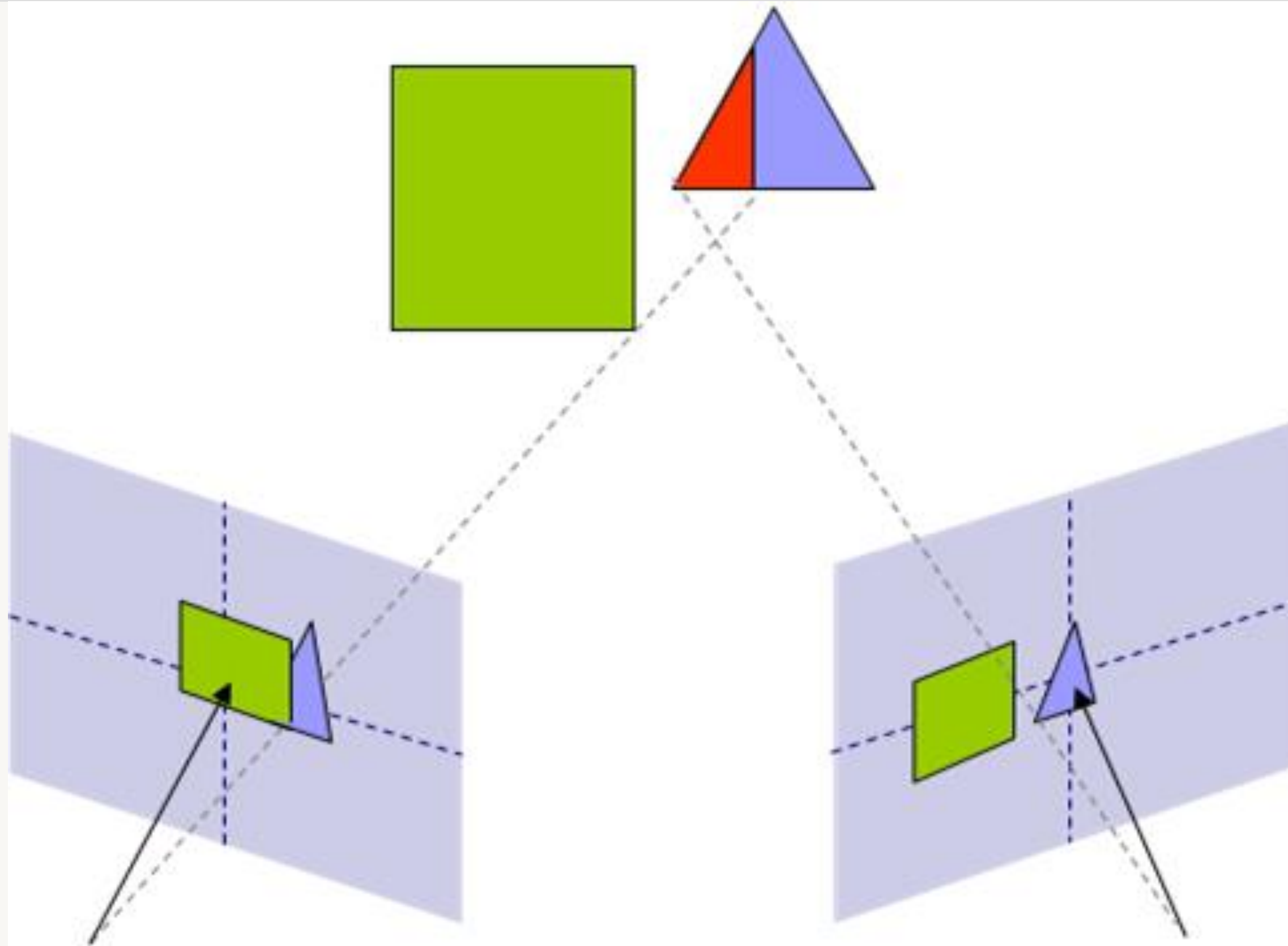
Eye Strain



Accommodation Vergence Conflict



Binocular Occlusion Conflict



Flicker



Refresh rate of the display

Flashing lights

Scene Darkness

Photic seizure

Driving by a picket fence ~ Flashing stimuli

1 ~ 50 Hz

0.01% of the population

Aftereffects



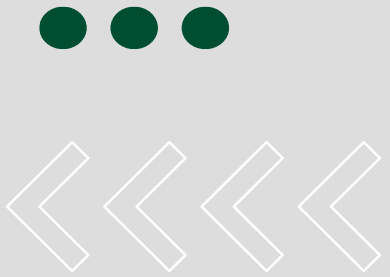
**Perceptual instability of the world, disorientations,
Flashbacks**

10%, 1~2 hours

Alcohol intoxication

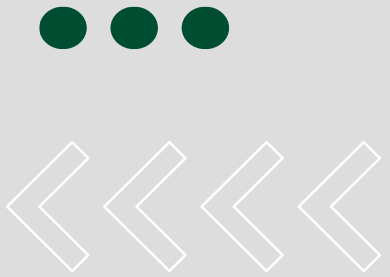
Aftereffects

Readaptation



- **Adaptation back to a real-world**
- **Field of view issues**
 - - adaptation and re-adaptation of wrong field of view
- **Drowsiness, disturbed locomotor and postural control, lack of hand-eye coordination**

Aftereffects



- **Natural Decay**
 - Close eyes and stay still
- **Active Re-adaptation**
 - Performing specific real-world task
 - E.g.) Hand-eye coordination
- **Dual adaptation**

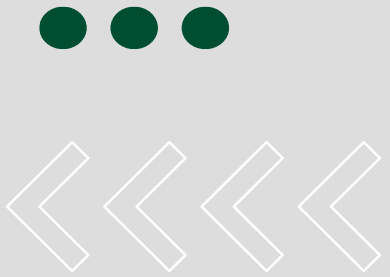
Hardware Issues



Physical Fatigue



Physical Fatigue



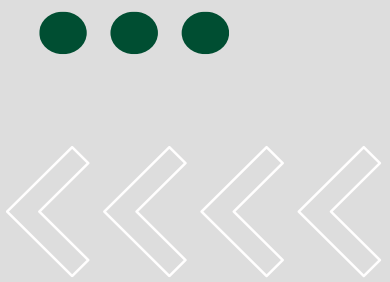
- **Center of Mass**
- **Gorilla Arm Syndrome**
 - Typically starts 10 seconds having to hold arms above the waist (without the rest)
 - Redesign hardwares and interactions
- **Real walking and standing**



Physical Fatigue



Hygiene and Fomite : Headset Fit



Slippage
Ergonomics design
Pump-inflatable liners

Injury



Physical trauma
Repetitive strain injuries
Noise-induced hearing loss

Summary



- **Health effects of VR**

- Sickness
- Eye strain
- Seizures
- Aftereffects and Readaptation
- Fatigues
- Injury
- Hygiene

**Need to design -
Feedback Contents**

Hardwares

Re-adaptation Procedures

Safety guards

Homework



- **Design an interaction mechanism**
 - To provide maximum immersive
- **Identify possible health issues in your design**
- **Discuss about possible conflict between immersive level and safety/healthy**

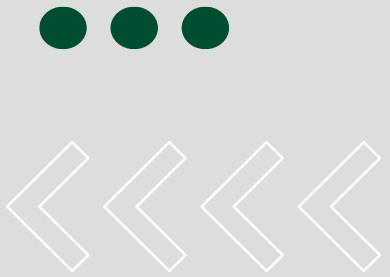


Latency and Sickness

HyungSeok Kim
Konkuk University



Introduction

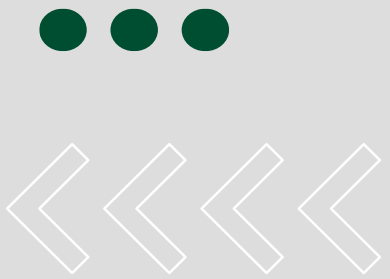


Fundamental Task The primary goal of a VR system is to maintain a stable scene that remains fixed in space despite the user's head movements, ensuring experiential fidelity.

The Issue: Latency

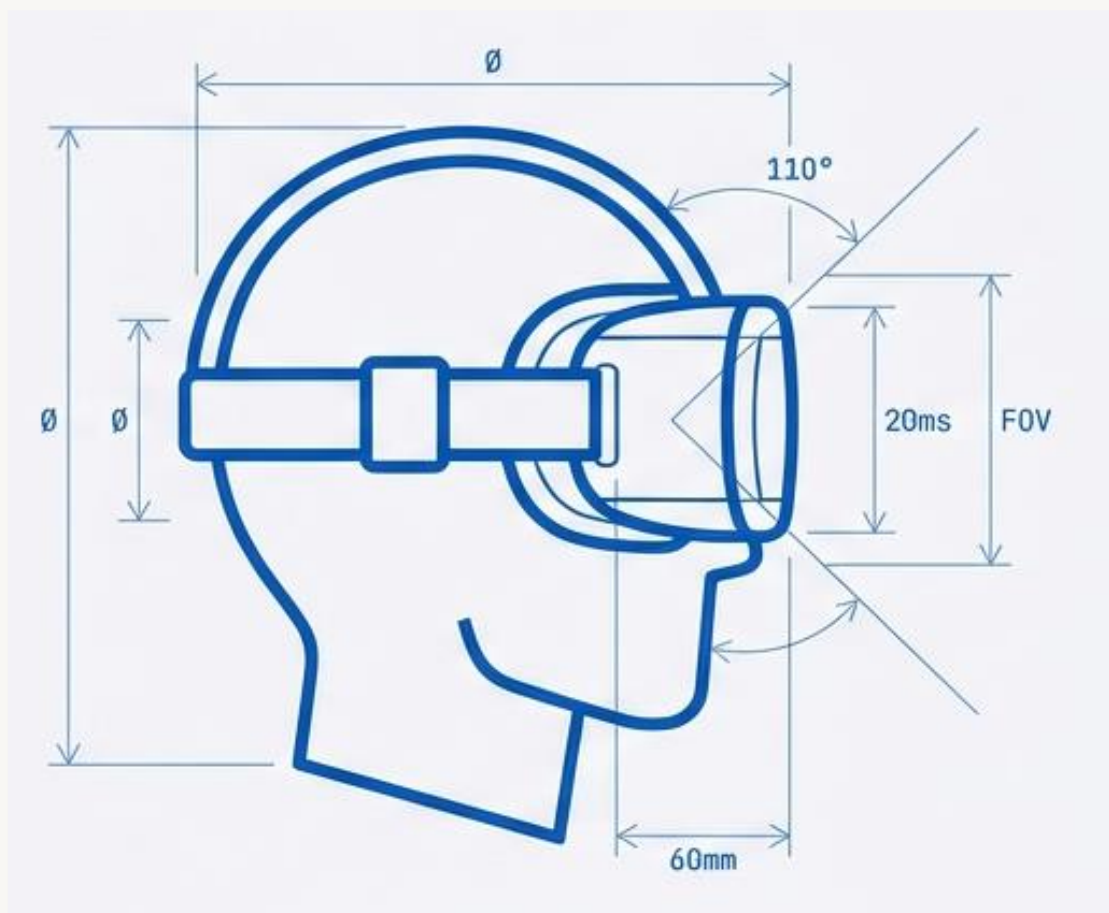
- **Definition:** The actual time elapsed from the start of a user's movement to the corresponding update of pixels on the display.
- **Variable Factors:** It is heavily influenced by display technology, including varying response and transition times.
- **Key Objective:** The ultimate mission is to eliminate spatial instability and stabilize the scene within real-world space to prevent unintended motion.

Negative Effects of Latency

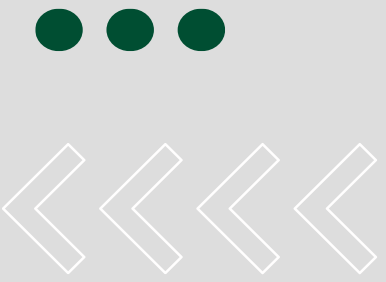


Consequences of latency

- Users don't perceive latency below 100ms.
- Static visuals appear unstable during head movement.
- Latency lags visual cues, causing sensory conflict and VIMS.
- Serious usability issues; major VR motion sickness contributor.
- Essential for VR creators to understand and minimize latency.



Negative Effects of Latency

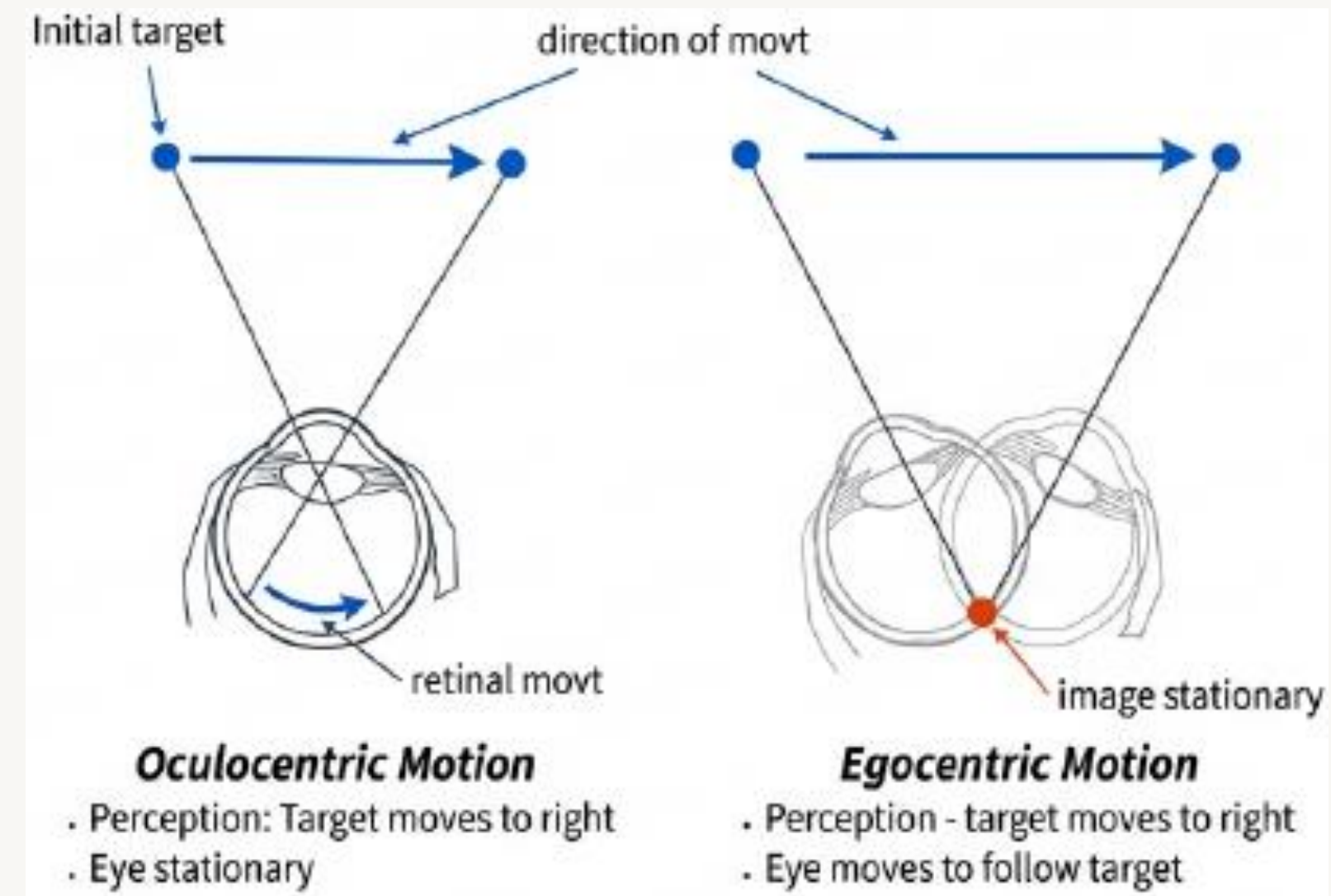


1) Degraded Visual Acuity:

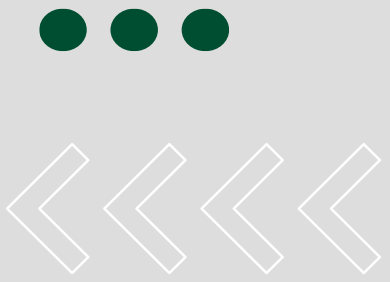
- Retina motion blur >2-3% reduces acuity.
- Sinusoidal head motion (0.5Hz at 20° or 130ms latency) peaks blur.

Retinal Image Slip:

- Lagging image causes ~4.8% slip in scene.
- Users' eyes stabilize in space, resulting in slip.



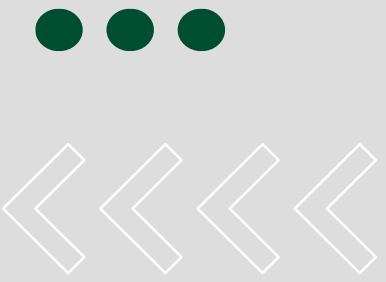
Negative Effects of Latency



2) Degraded Performance:

- Latency thresholds harming performance differ from perceptible levels.
- HMD studies show latency hinders operator learning.
- Head-tracking tasks fail to improve with training at $\geq 120\text{ms}$ latency.
- Subjects cannot compensate for latencies in such tasks.

Negative Effects of Latency

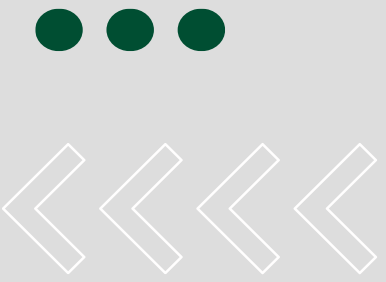


3) Break-in-Presence:

- Latency reduces sense of presence.
- With head movement, scene motion mismatches real world.
- Incorrect motion distracts users, breaking virtual immersion.
- Reveals simulation as illusion.



Negative Effects of Latency



4) Negative Training Effects:

- Unintended performance drop after task training.
- Latency causes negative training outcomes.
- Seen in desktop displays.
- Also in large-screen driving simulators.

Cunningham et al. 2001b

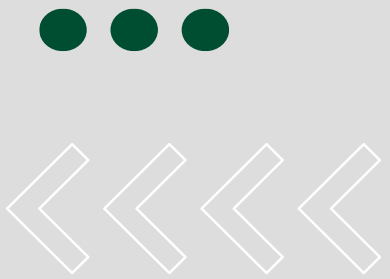
Driving in the Future: Temporal Visuomotor Adaptation and Generalization
[Cunningham et al. 2001b]



Movie 1. A demonstration of the experimental setup and task with a 430-millisecond delay.

Negative Effects of Latency

Latency Thresholds

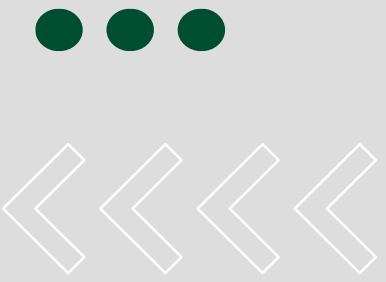


Low Latency:

- Engineers target "low latency" in HMDs without clear definition.
- Latency must be imperceptible during scene motion.
- Thresholds drop with greater head motion.
- Quick head movements make latency easier to detect.

Negative Effects of Latency

Latency Thresholds



Head Yaw Thresholds: NASA studies show 85ms variation due to bias, movement type, individuals; consistent ~4-40ms JND; no scene complexity effect; consistency key.

Head Turn Motion Thresholds: Measure 7.4ms (system delay); psychophysical model verifies decreased sensitivity with head motion as latency-induced slip outpaces scene motion.

Full VR Application: Optical see-through displays need lower thresholds since users discriminate real vs. synthesized cues directly.



Negative Effects of Latency

Delayed Perception as a Function of Dark Adaption



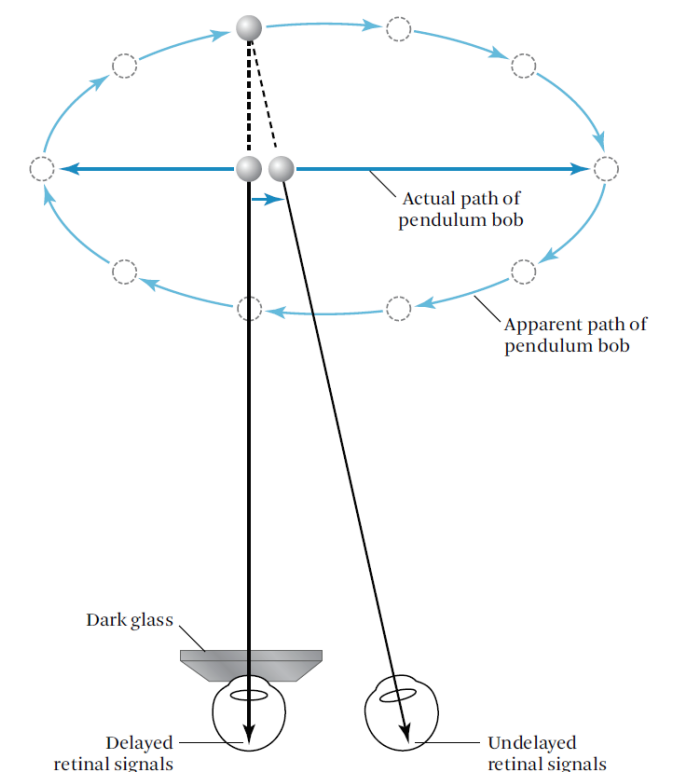
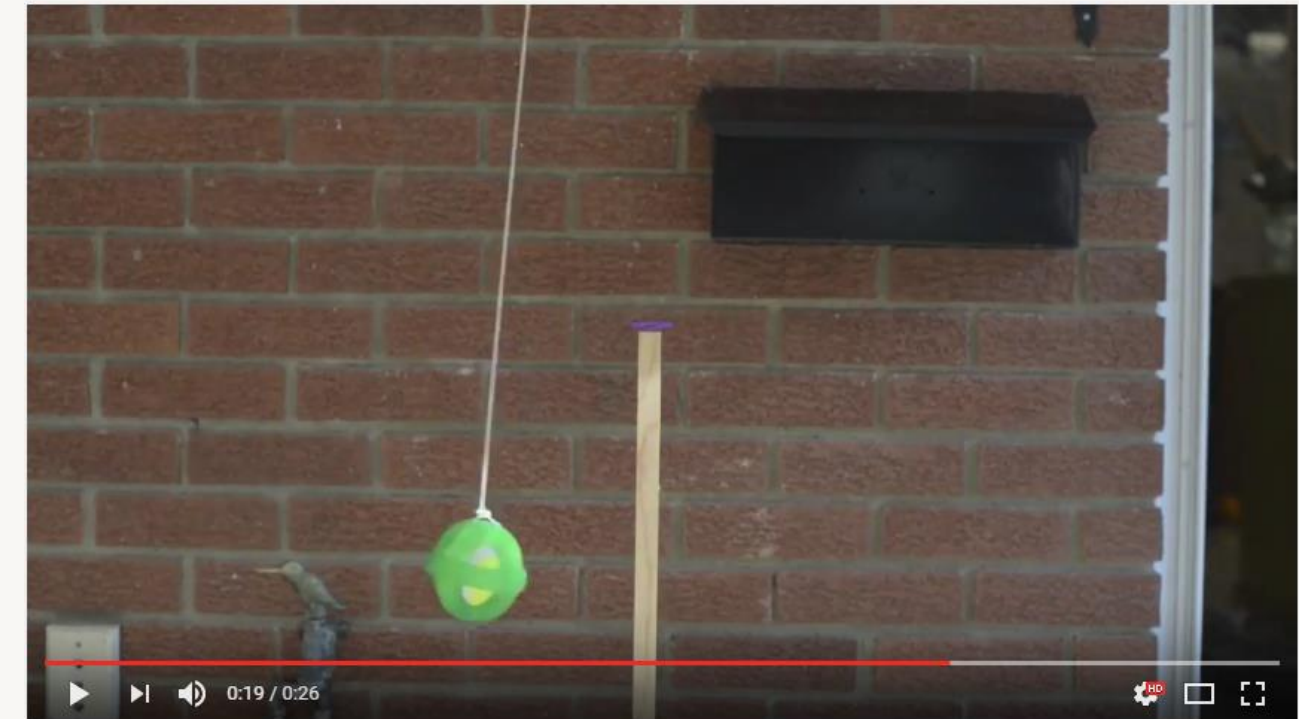
Light/Dark Delay: Brain may recognize darker stimuli/calibrate delay; imprecise dark/low-intensity delay causes motion smear, localization errors; observers bias stable objects in dark.

2nd Possibility: Brain adapts latency-HMD link lifetime; correlates dark adaptation/stimuli intensity to delay; expects more delay with HMD/sunglasses, unlike real dark delays.

Pulfrich Pendulum Effect: Depth illusion from unequal dark adaptation per eye; pendulum swings in ellipse (not flat arc).

Core Question: Why no instability/motion sickness in real dark/sunglasses delays vs. HMD?

<http://youtu.be/5Hh3HT8XUmw>



The Pulfrich penulum effect [Gregory [1973]]

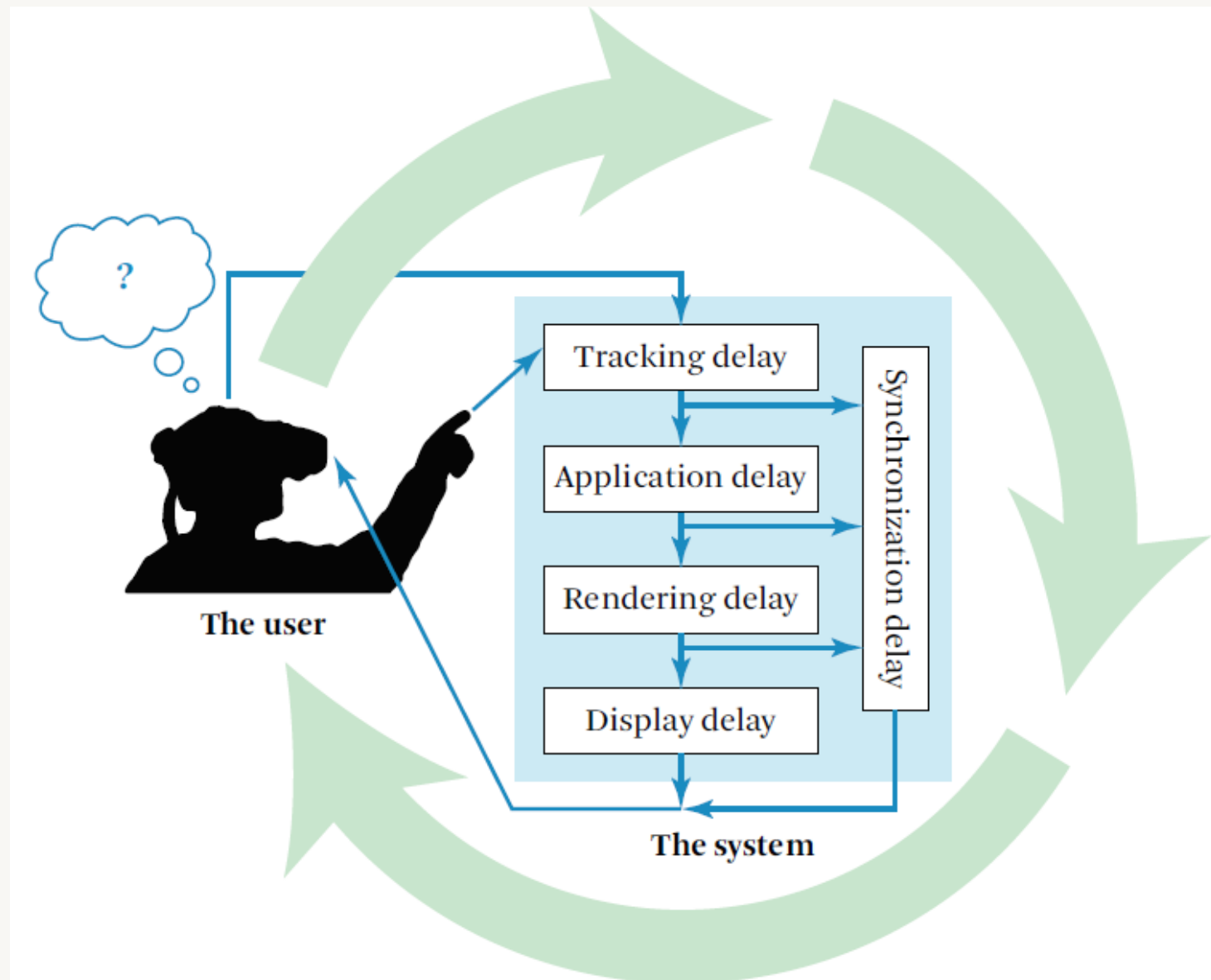
Negative Effects of Latency

Sources of Delay



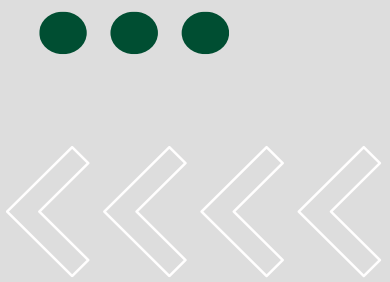
* System Delay

- The sum of delays from tracking, application, rendering, display, and synchronization among components
- The system delay is greater than the inverse of the update rate



Negative Effects of Latency

Sources of Delay



Sources of Delay in VR/HMD Systems

Tracking Delay: Time from body movement to tracker sensor input; includes filtering techniques complicating analysis; network delay as part.

Application Delay: Time from tracking data receipt to world stage update; computes user interactions; varies by task complexity; often asynchronous (e.g., weather sims).

Rendering Delay: Time from new data entering graphics pipeline to resulting image; depends on scene complexity, quality, passes, HW perf; frame rate inverse; controllable by devs.

Display Delay: Time from graphics card signal to pixel intensity change; varies by HMD display tech.

Negative Effects of Latency

Sources of Delay





Thankyou!



번호



이메일

